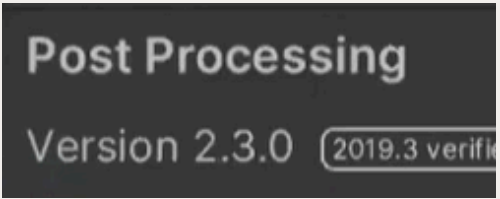
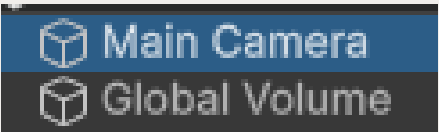


URP/HDRP 프로젝트에서의 포스트 프로세스 볼륨 사용법

<기본 제공 Global Volume 사용>



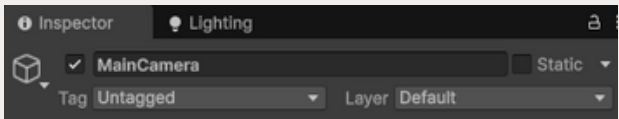
PostProcessing 설치



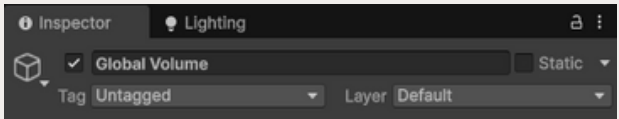
MainCamera
>브레인 필요X



URP/HDRP에서는 기본 Global Volume이 제공.

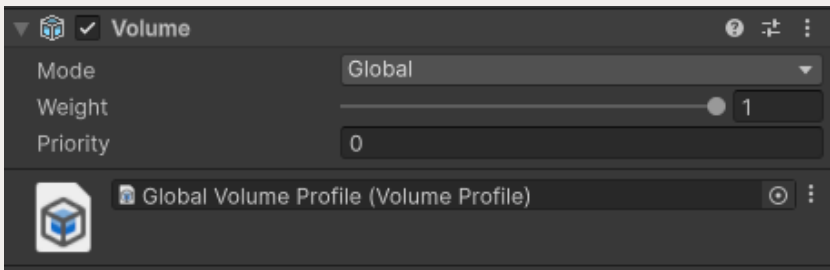


Layer : Default



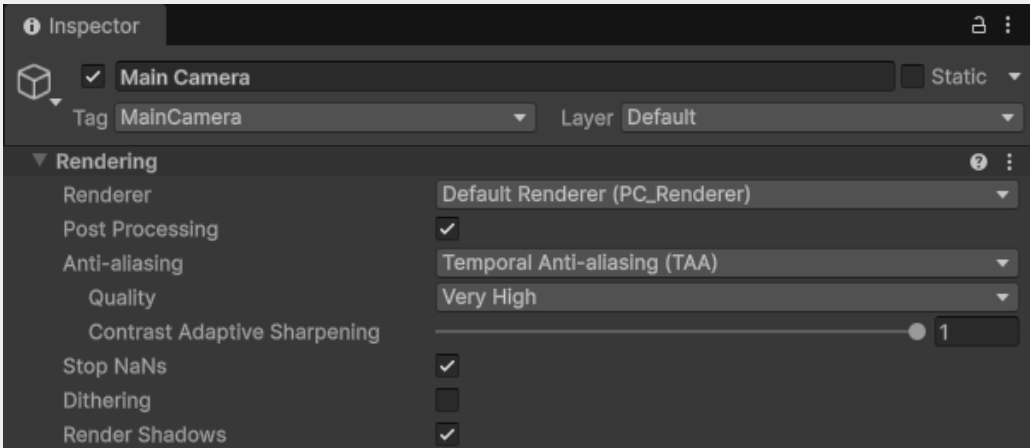
Layer : Default

<새로 Volume 생성시>



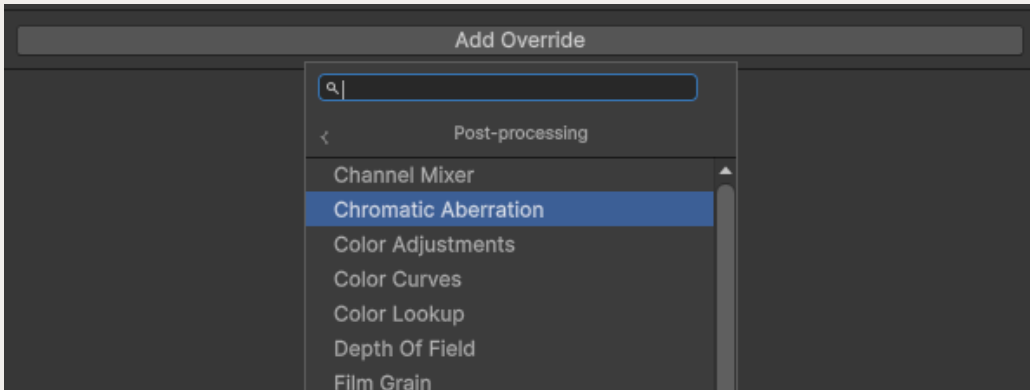
하이어라키>Volume>Global
>Local(Box/Sphere)
>Volume Profile 생성

<메인카메라 기본 설정>



• Camera>PostProcessing[v]

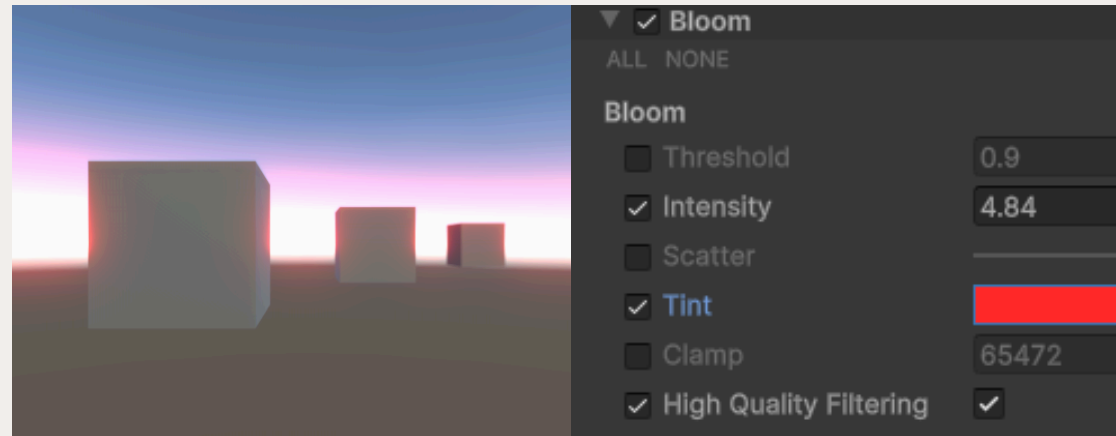
<PPV 사용법>



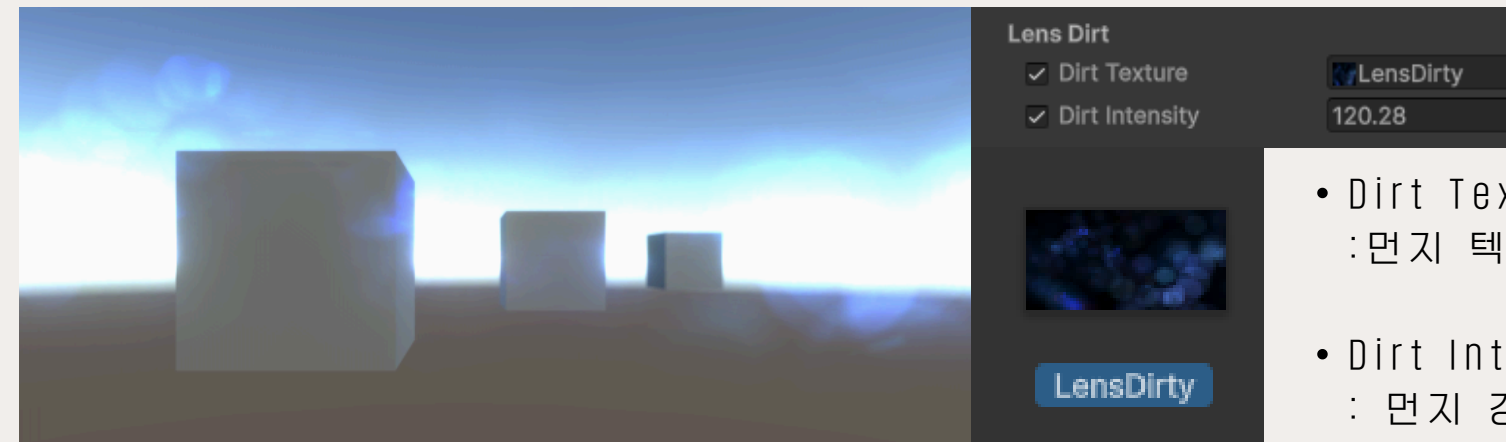
Add Override
> 볼륨 옵션 추가후 사용
(블룸, 컬러그레이드 등등)

<대표적인 포스트프로세스 볼륨 기능>

• Bloom(Bloom/Lens Dirt)

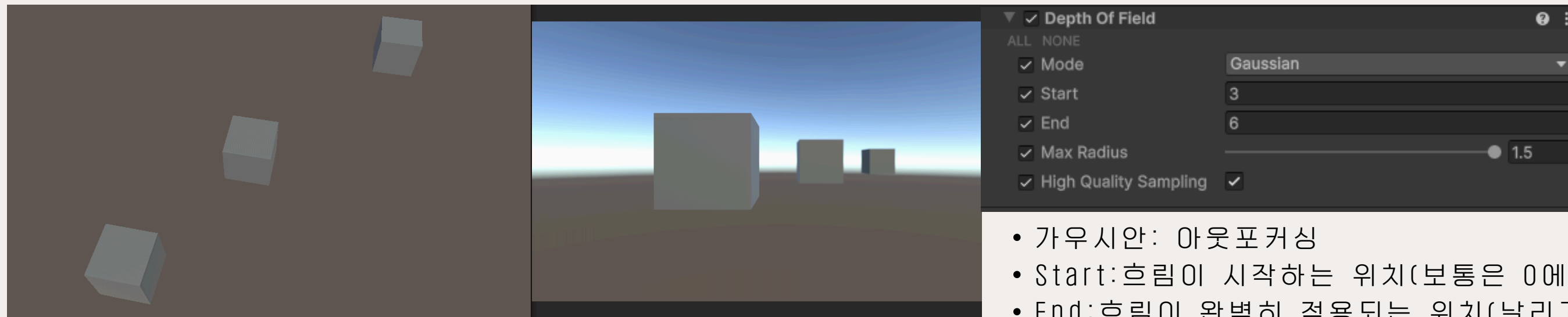


- Intensity
: 블룸강도
- Tint
: 블룸색상
- H.Q.F
: 블룸 품질



- Dirt Texture
: 먼지 텍스처
- Dirt Intensity
: 먼지 강도

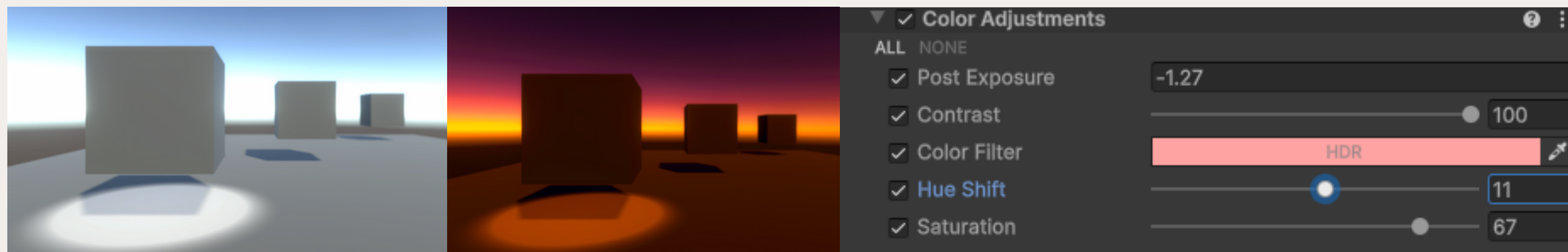
• Depth of Field(아웃포커싱)



- 가우시안: 아웃포커싱
- Start: 흐림이 시작하는 위치(보통은 0에 두고)
- End: 흐림이 완벽히 적용되는 위치(날리고픈 오브젝트 위치=End)
- Max Radius: 흐려짐 정도(높을 수록 많이 날림, 3 이상이 적당)

카메라:0(Z) / 1번큐브: 3(Z) / 2번큐브:6(Z) / 3번큐브:10(Z)

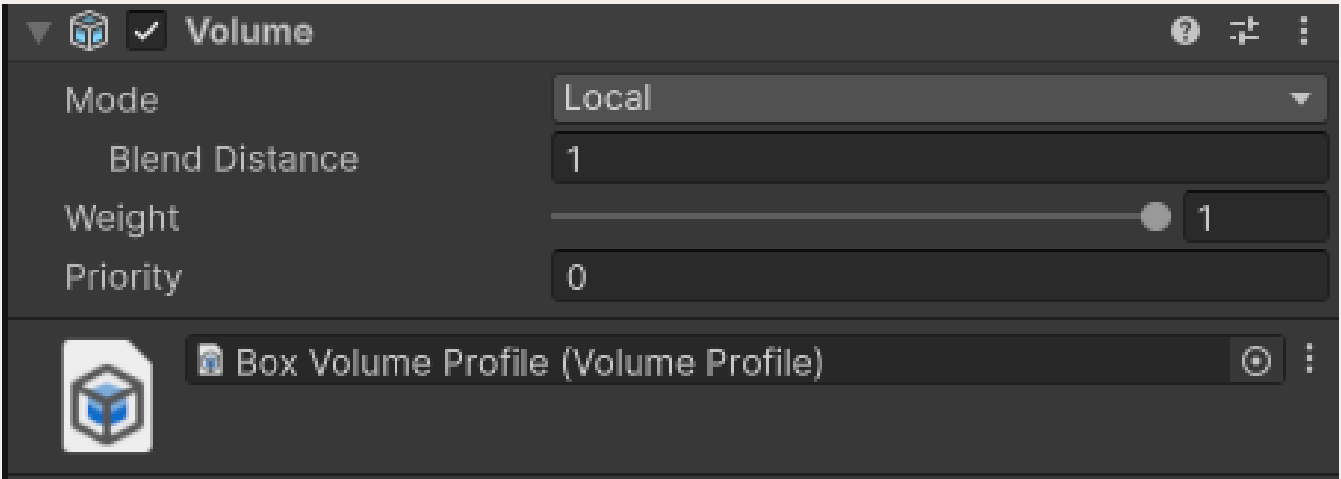
• Color Adjustment(색보정)



- PostExposure: 노출(밝기)
- Contrast: 명암 대비
- Color Filter: 색조 입히기
- Hue Shift: 색조 변경
- Saturation: 채도

<Box Volume 사용>

<Blend Distance 존재 상황>



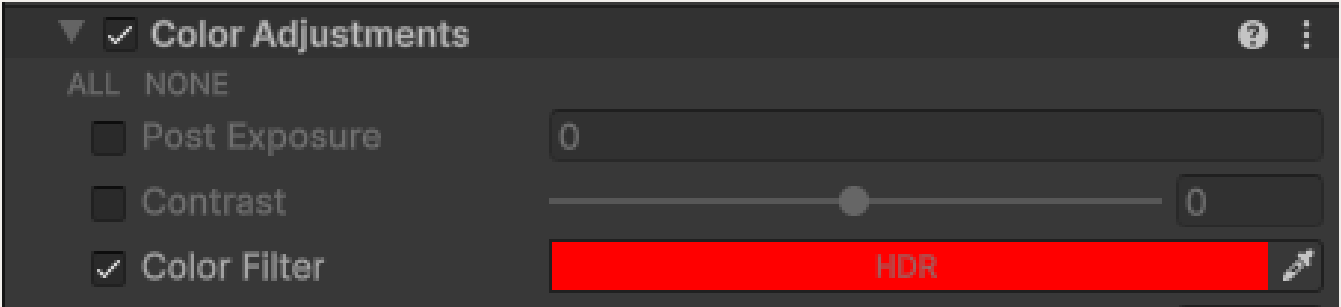
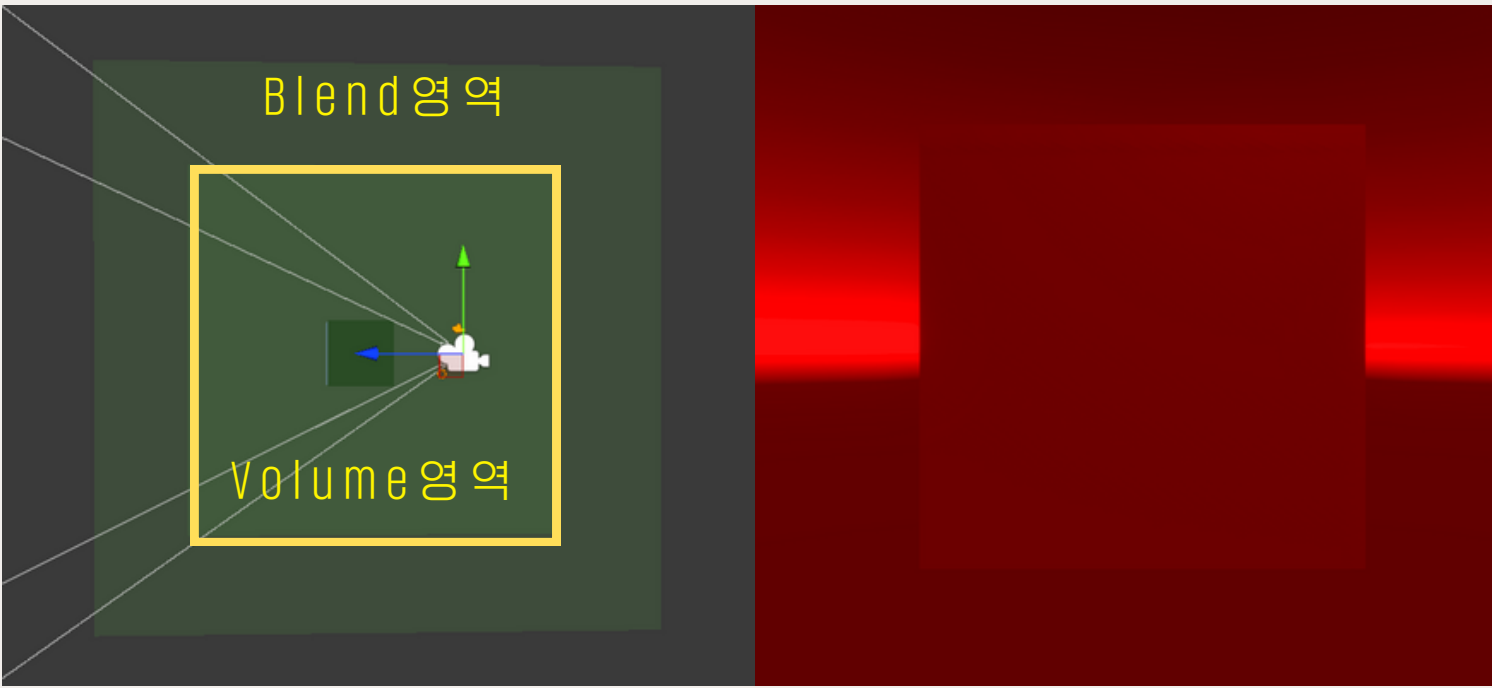
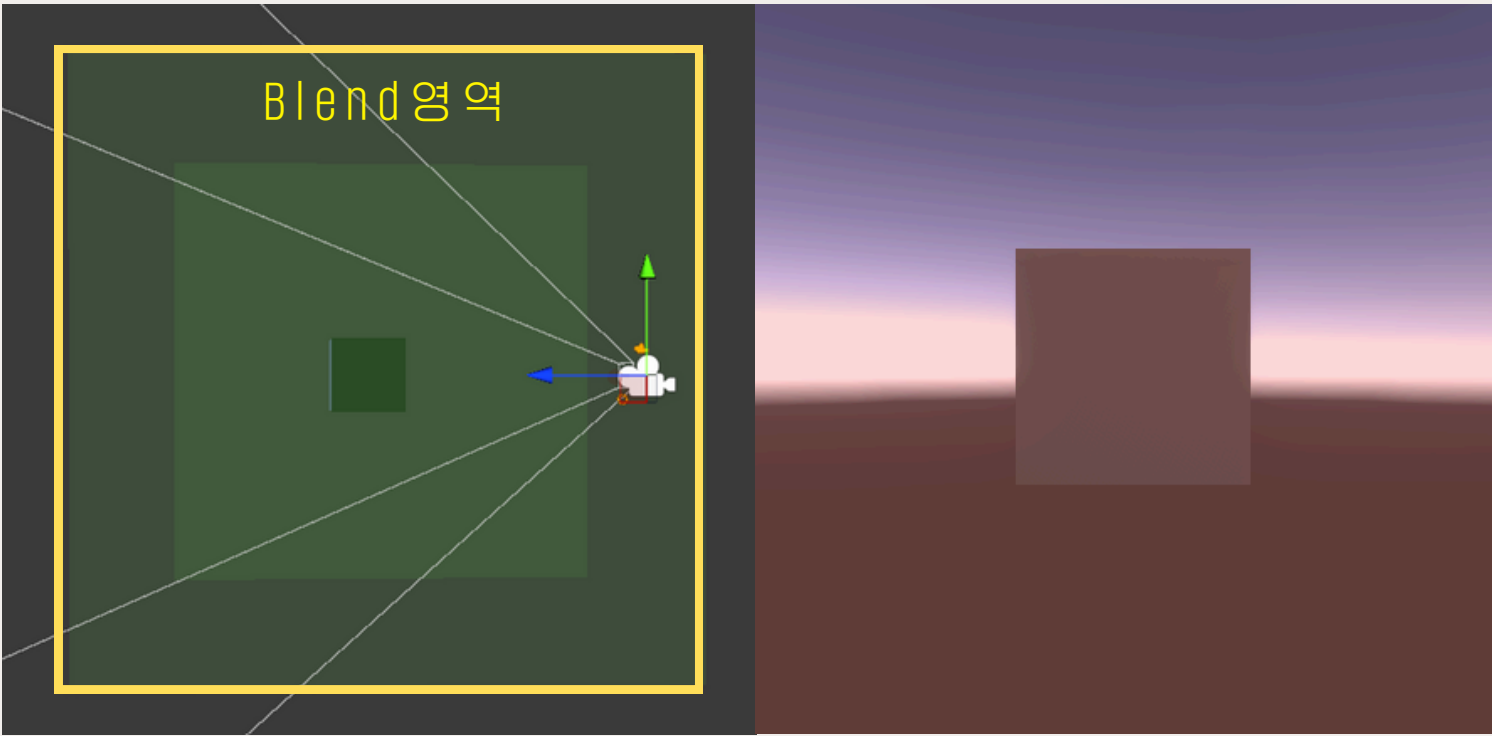
Volume Profile 생성

> Mode : Local(고정)

> Blend Distance(블랜드 영역 너비)

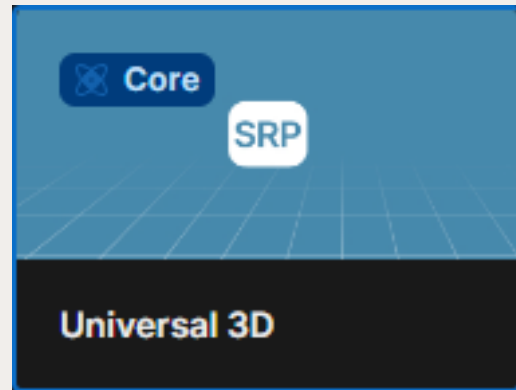
* 0 = 박스 볼륨 내에서만 적용

* 0 < 블랜드 되는 중간 지점 존재

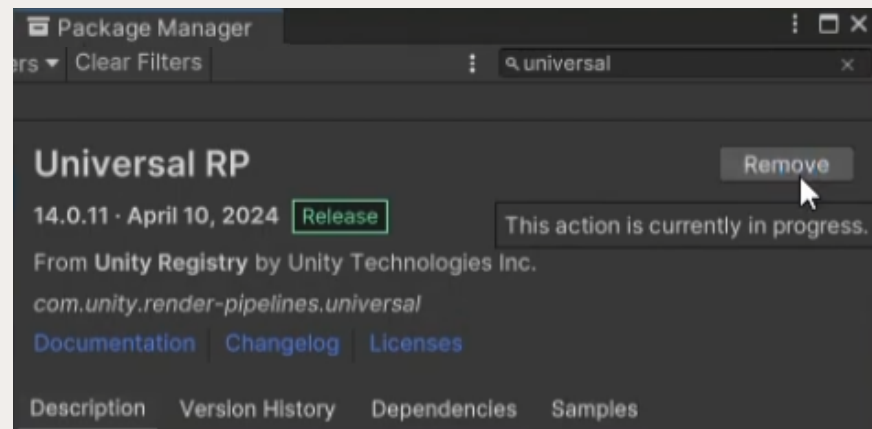


포스트프로세스 볼륨 기능 추가

<URP에서 Realistic Lighting 설정방법>



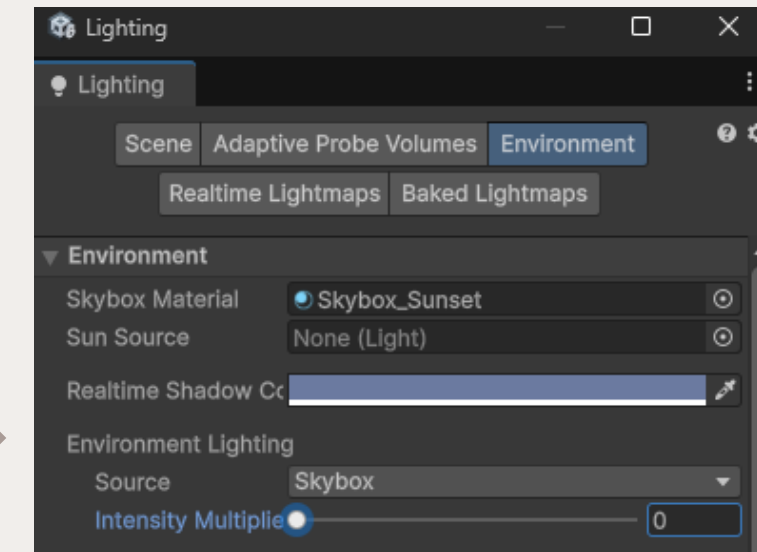
URP 프로젝트 생성



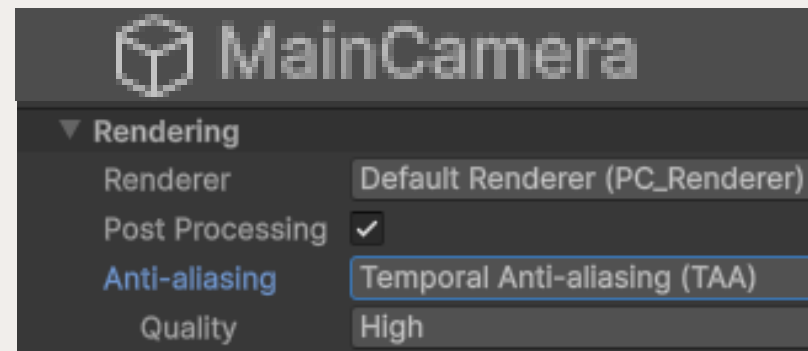
패키지 매니저 > URP 설치 확인



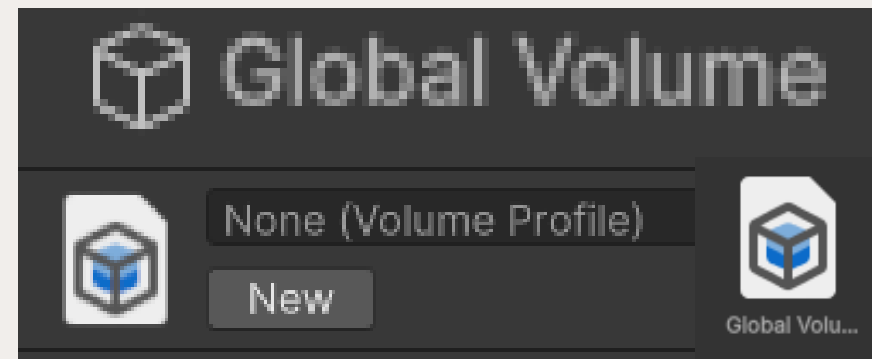
패키지 임포트 > 씬 열기
(Country Yard with Statue)



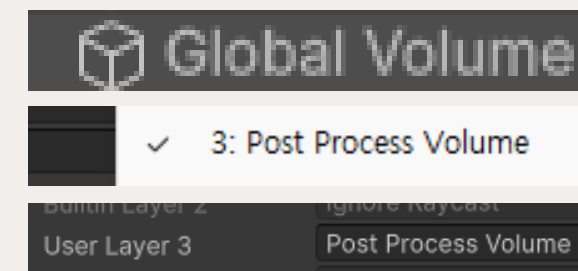
Window > Rendering >
Lighting > Environment
> Intensity Multiplier: 0
(기본 간접광 설정을 비활성화)



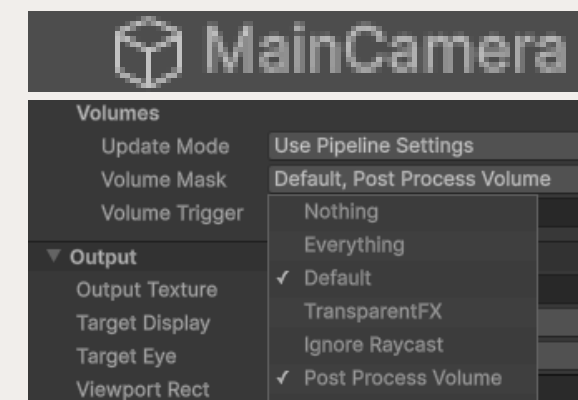
MainCamera 생성(Camera)
> 포스트프로세싱 [v]
> 안티에일리어싱: TAA/High
(렌더링 계단 현상 제거)



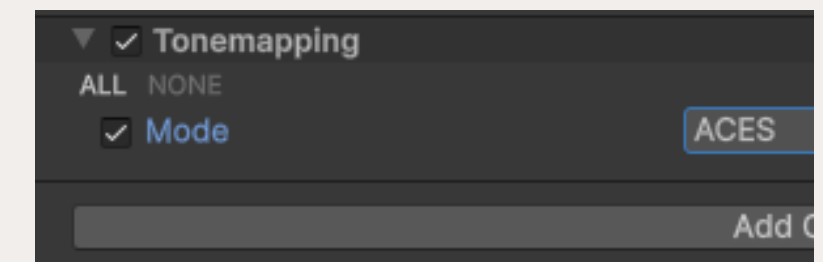
글로벌 볼륨 생성 > New로 볼
륨데이터 에셋 생성



글로벌 볼륨 > 레이어 생성
> Post Process Volume

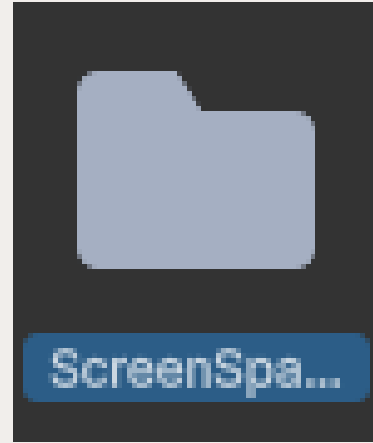


메인카메라 > 볼륨 마스크
> Default+PPV 선택

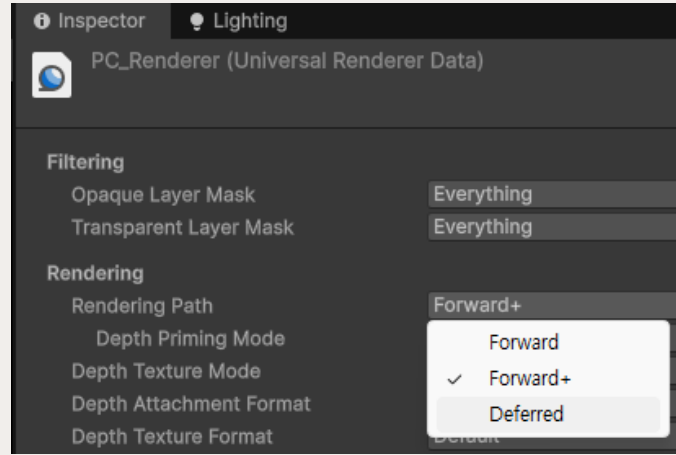


글로벌 볼륨 > Add Override
> Tonemapping > Mode: ACES
(영화 산업 표준 컬러 매핑 시스템)

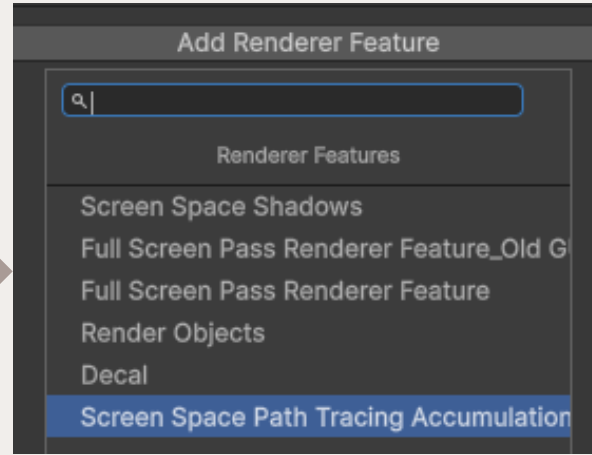
<URP에서 Realistic Lighting 설정방법>



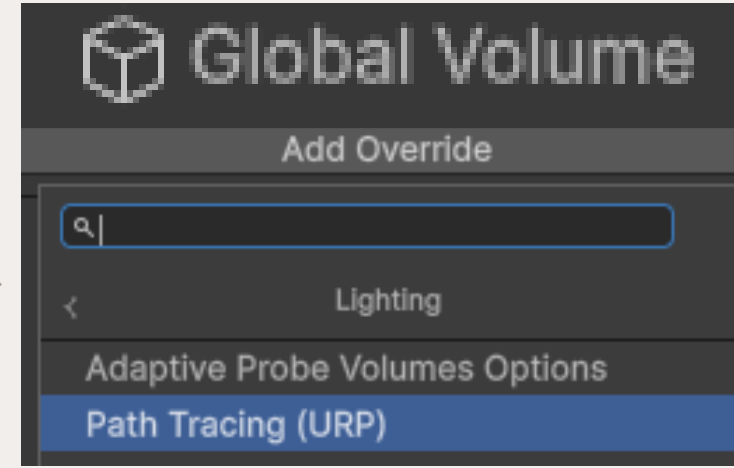
SSPT패키지 임포트



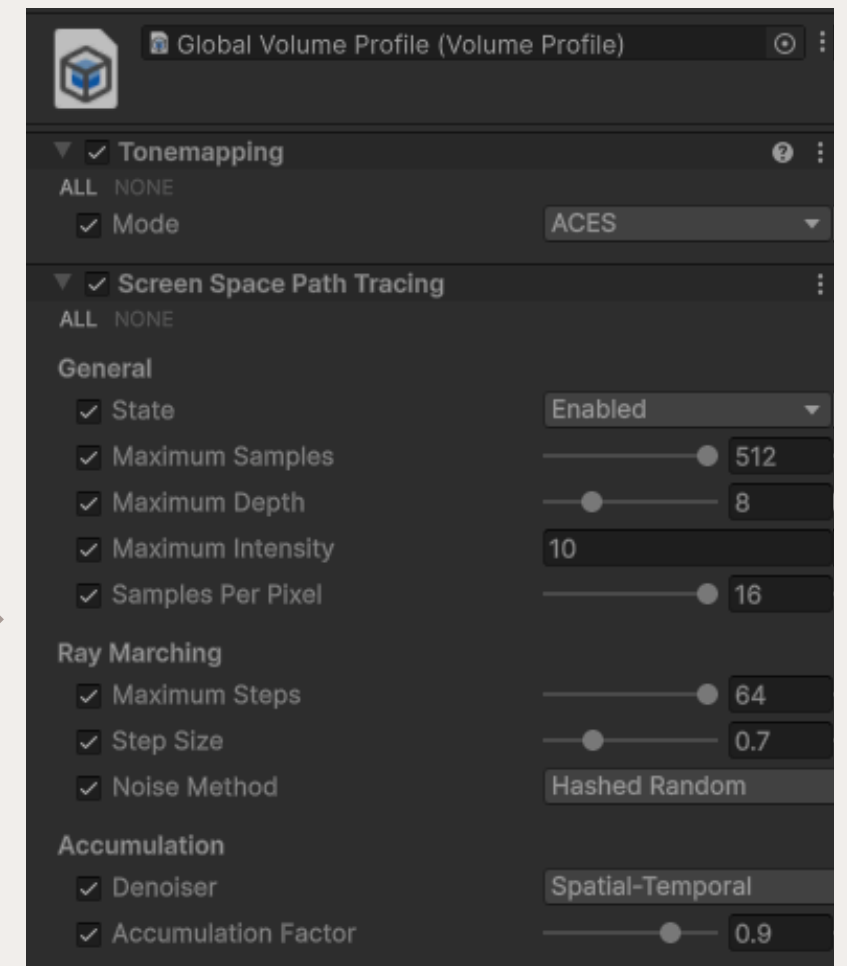
Settings > PC_Renderer
> Render Path: Deferred



>Add Renderer Feature
>SSPT추가



>글로벌 볼륨 > Add Override
>Lighting > SSPT추가



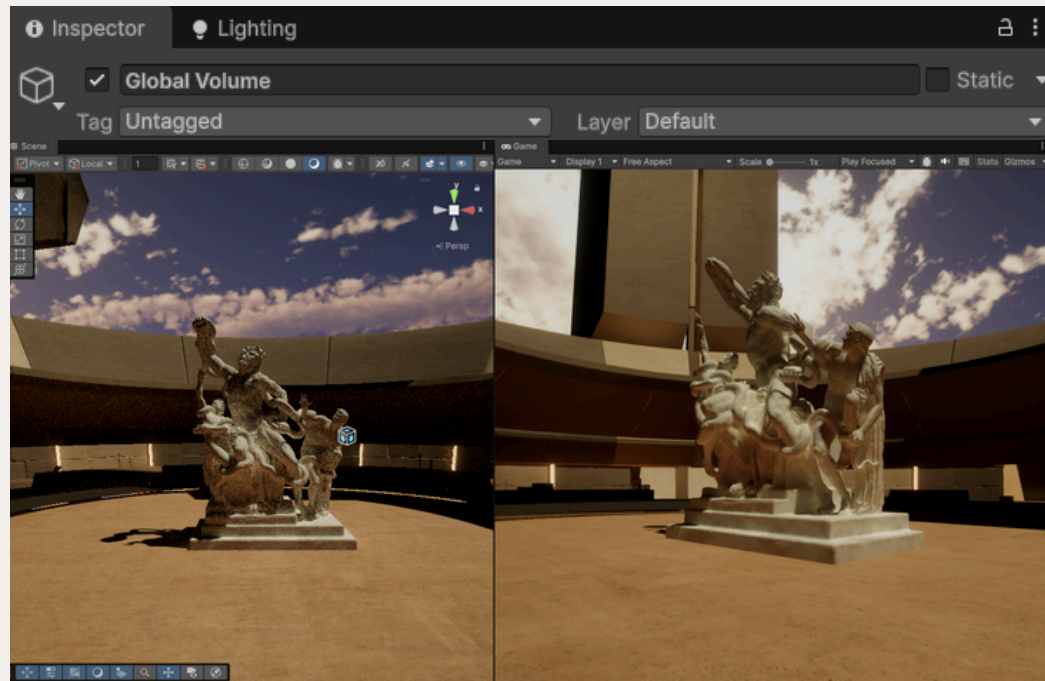
General

- >State: Enabled(활성화)
- >Maximum Samples:샘플수 (512)
 - 높을수록 고품질
 - Cinematic:256 ~ 1024
 - Real-time:64 ~ 256
- >Maximum Depth:빛 반사 횟수(8)
 - 높을수록 정확한 간접광
- >Maximum Intensity:빛 최대 밝기 Clamp(10)
- >Samples Per Pixel:픽셀당 샘플(16)
 - 높을수록 노이즈 감소

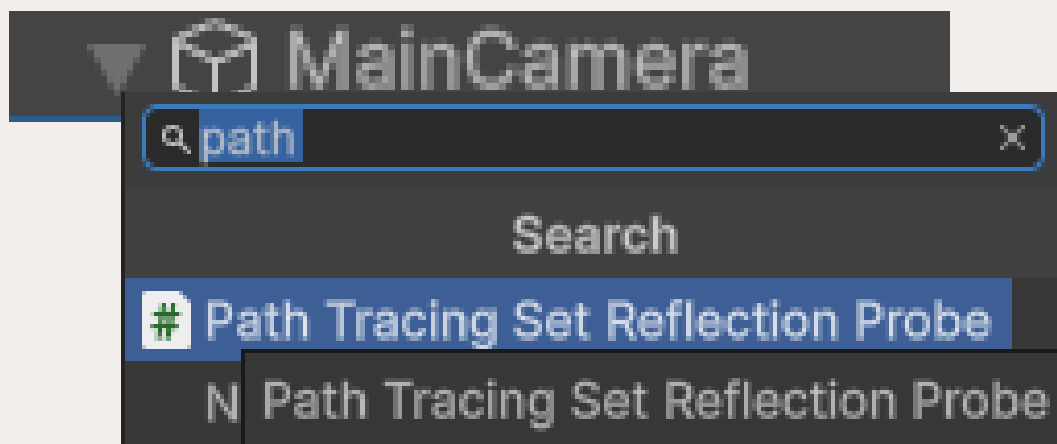
Ray Marching(레이트레이싱)

- >Maximum Steps:단계 수(64)
 - Cloud / Fog / Volume 품질 영향
- >Step Size (0.7)
- >Noise Method:노이즈 분산 방식 (영상Hashed Random/게임BlueNoise)

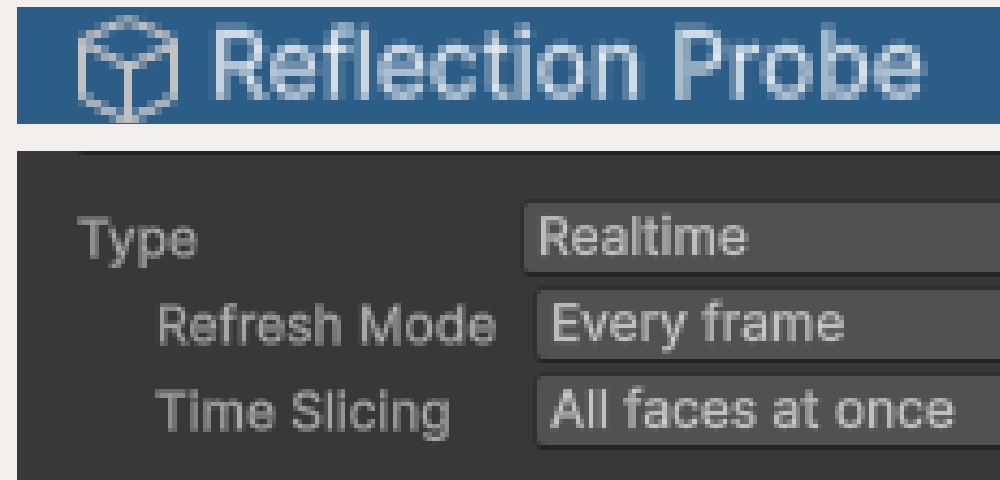
<URP에서 Realistic Lighting 설정방법>



Scene뷰에서도 적용상태 확인시>Global Volume의 레이어를 Default로 일치



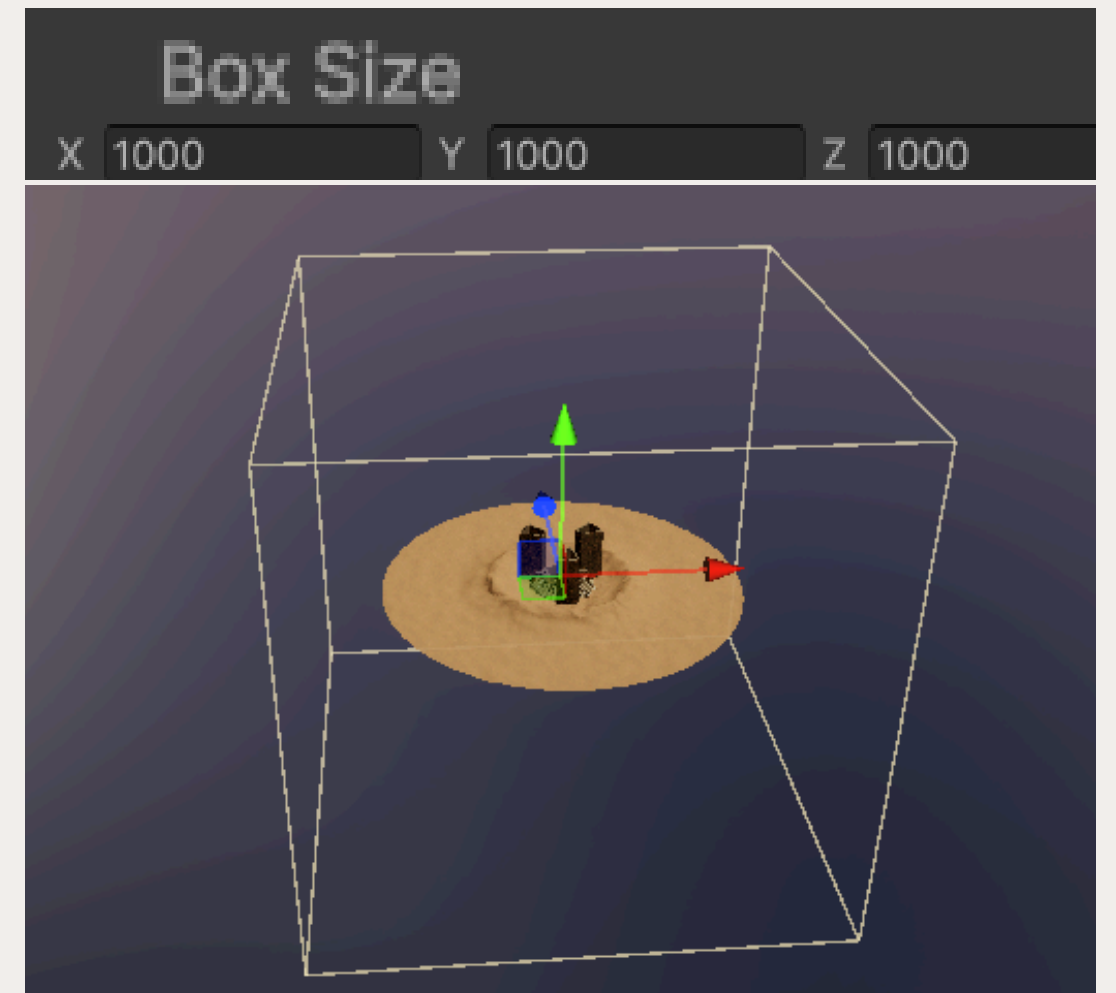
Main Camera > Add Component
> Path Tracing Set Reflection Probe



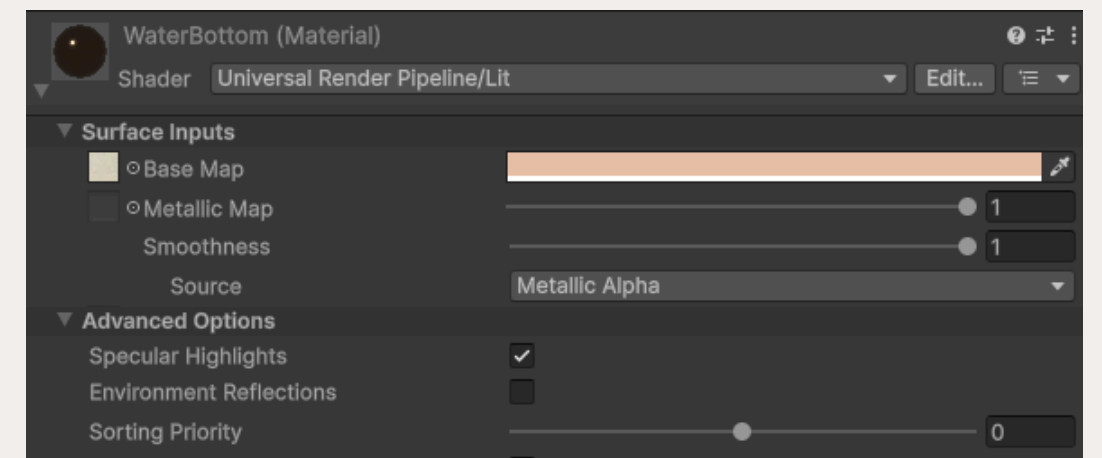
Hierarchy>우클릭>Lighting Reflection Probe 생성>Type:Realtime > Refresh Mode:Every frame



Main Camera 하위로 Reflection Probe 종속
> Reflection Probe > Culling Mask: Every

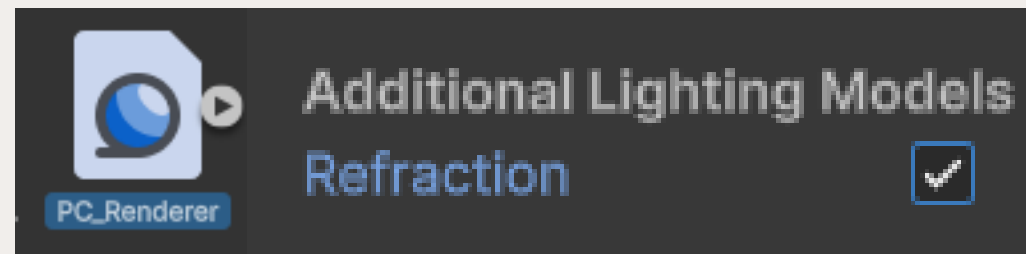


Probe Box Size: 공간을 다 덮을 만큼



바닥 오브젝트 > 머티리얼 > 메탈릭 ↑, 소
프트니스 ↑ > Environment Reflections
[] 비활성화: 자체 반사기능 끄기

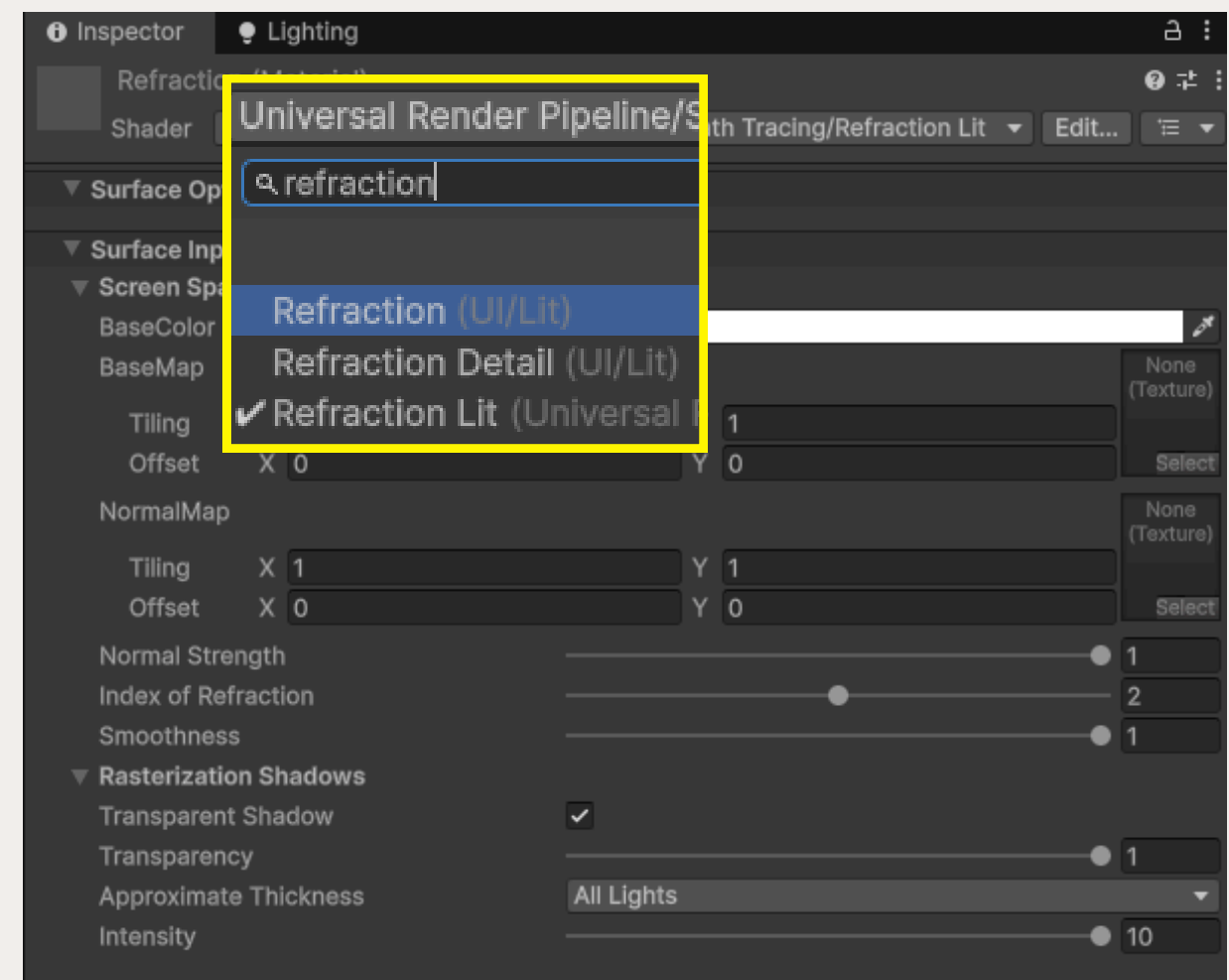
<URP에서 Realistic Lighting 설정방법>



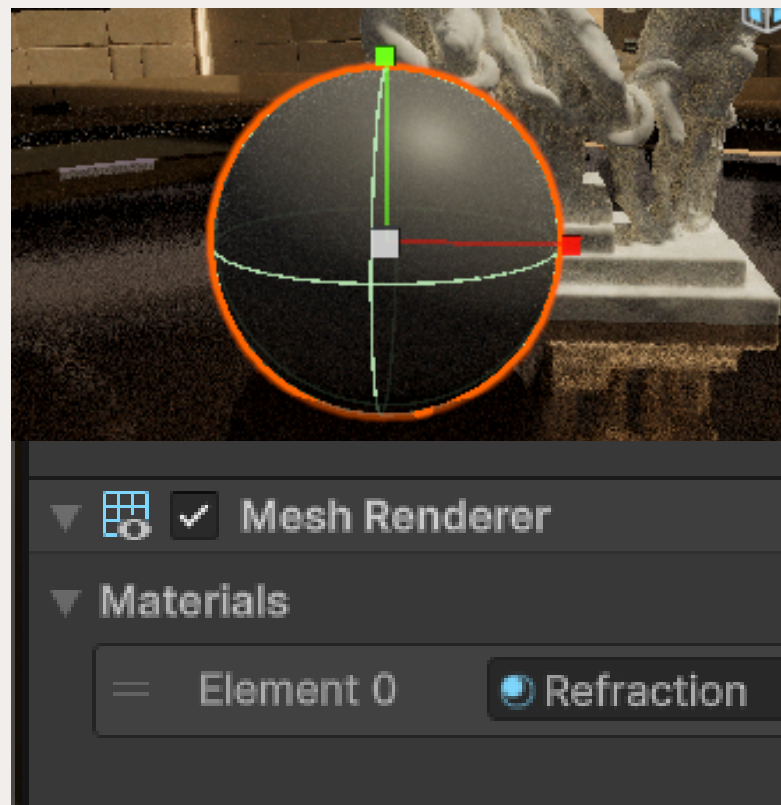
PC_Renderer > Additional Lighting Models[v]



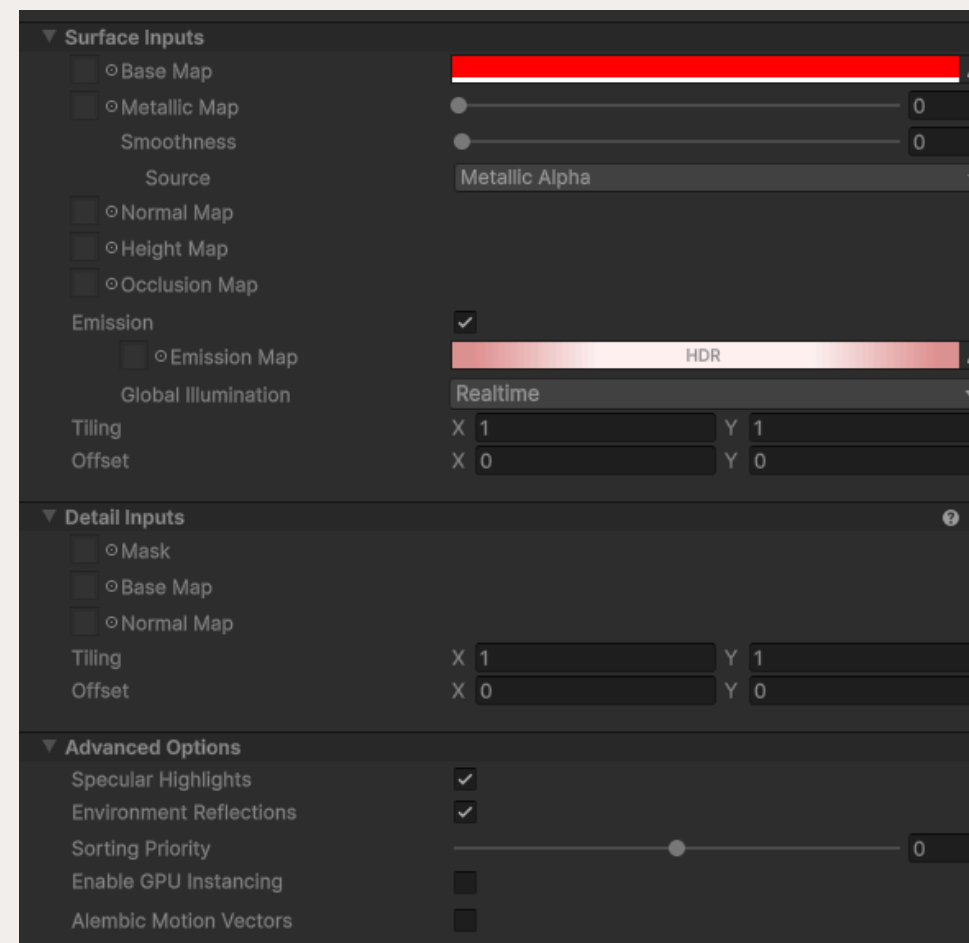
Refraction 머티리얼 생성



셰이더: Refraction Lit 선택 > 빛 굴절 머티리얼 옵션 설정



스피어 생성 > 머티리얼 > Refraction 머티리얼 할당

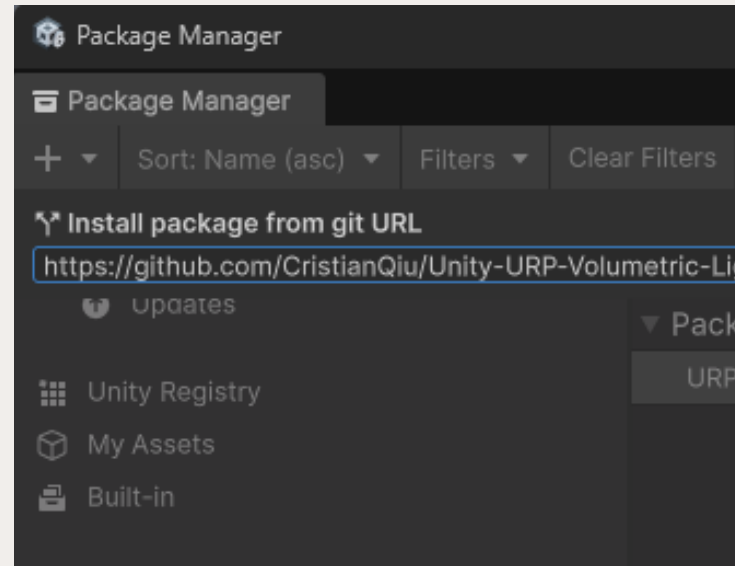


Emissive 머티리얼 생성 > 셰이더: URP Lit > 옵션 설정

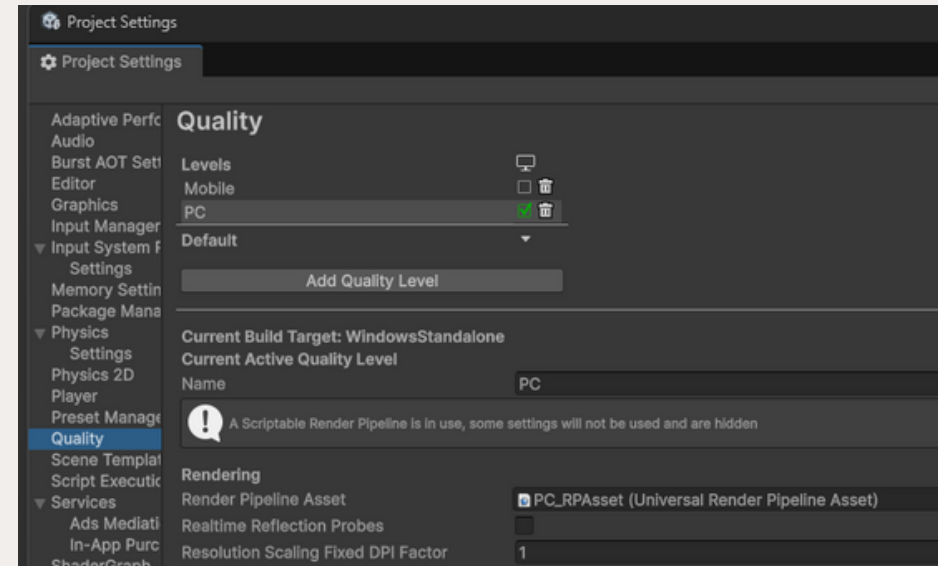


큐브 생성 > 머티리얼 > Emissive 머티리얼 할당

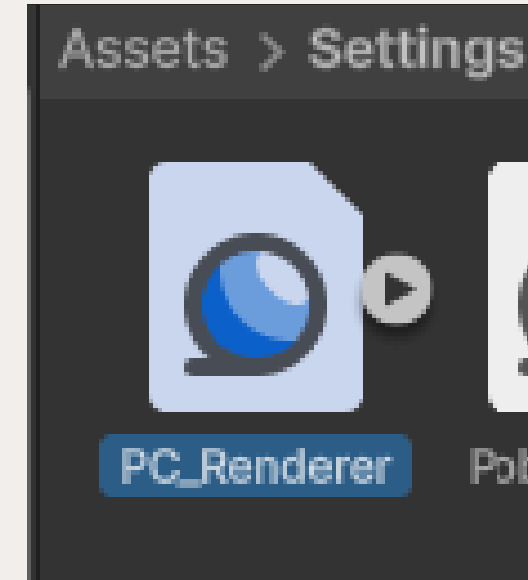
<URP에서 볼류매트릭 Fog 생성하기>



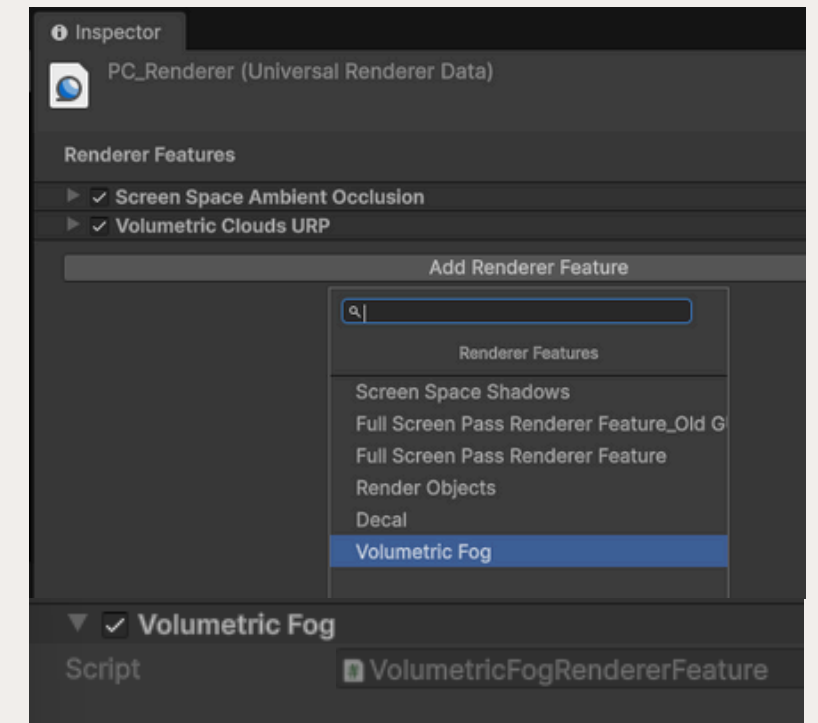
PackageManager>+
>Intall Pck from Github URL



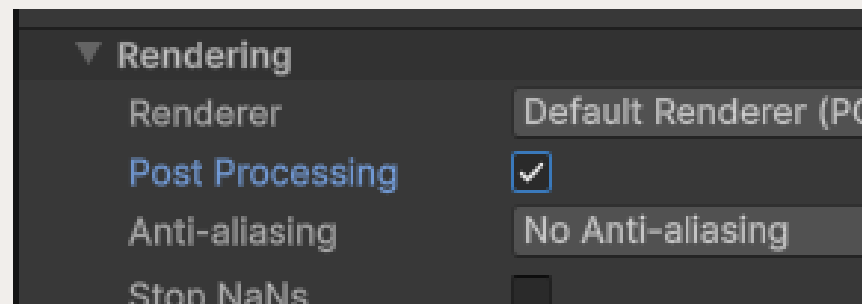
프로젝트 세팅 > Quality > Render
Pipeline Asset : URP 확인



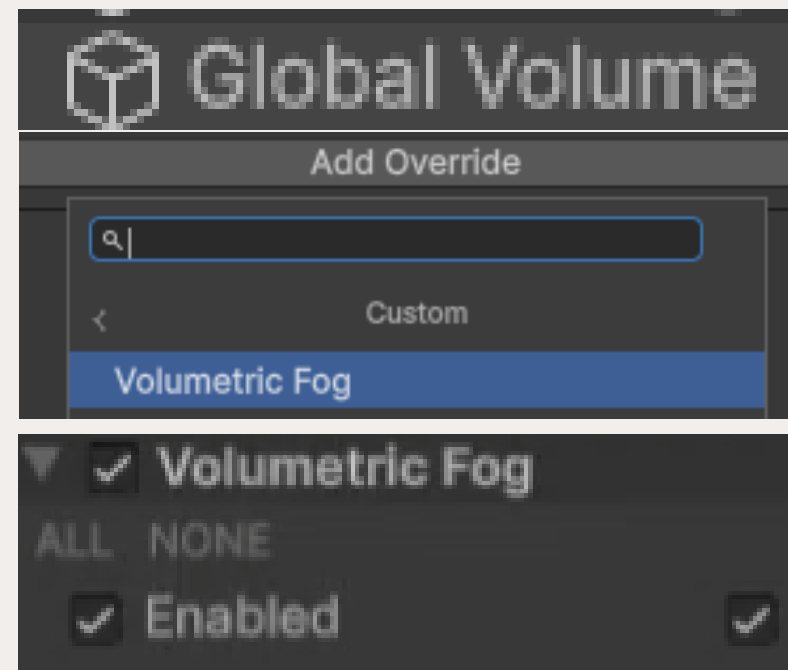
Asset>Setting
>PC_Renderer



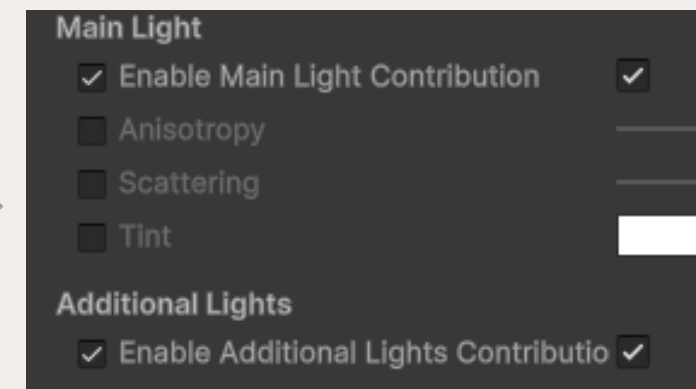
Asset>Setting>PC_Renderer
>Add Renderer Feature> URP Fog



MainCamera > Inspector
> PostProcessing[v]

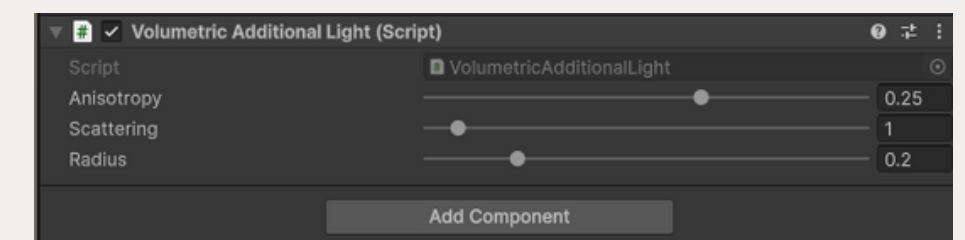
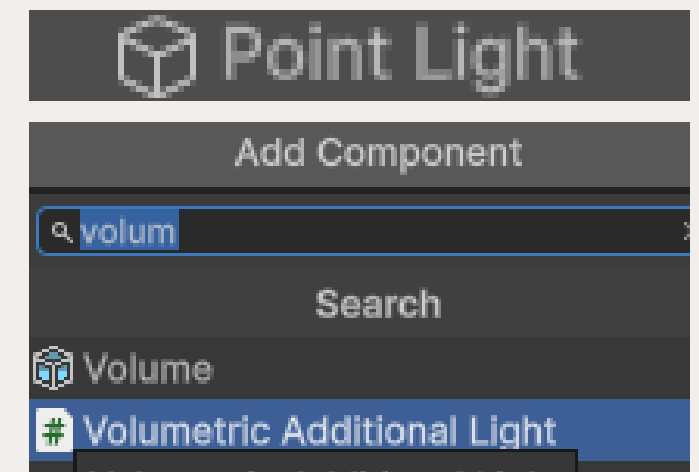


Hierachy > Global Volume
>Inspector > Add Override
> Custom > Volumatic Fog
>Enabled : [v]



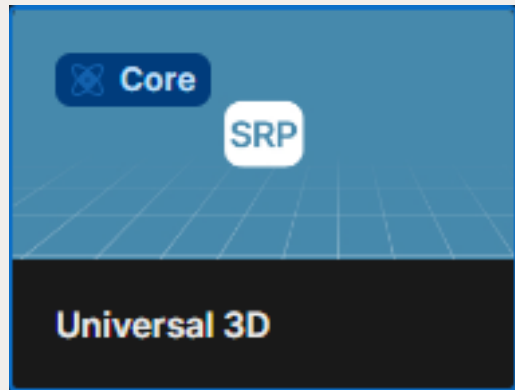
[v]Enable Main Light
Contributuion:디렉셔널 라이
트의 간섭 활성화

[v]Enable Additional Lights
Contribution: 라이트 오브젝
트의 간섭 활성화



- Anisotropy: 빛 진행방향
- Scattering: 산란 강도
- Radius:빛 영향 시각 범위

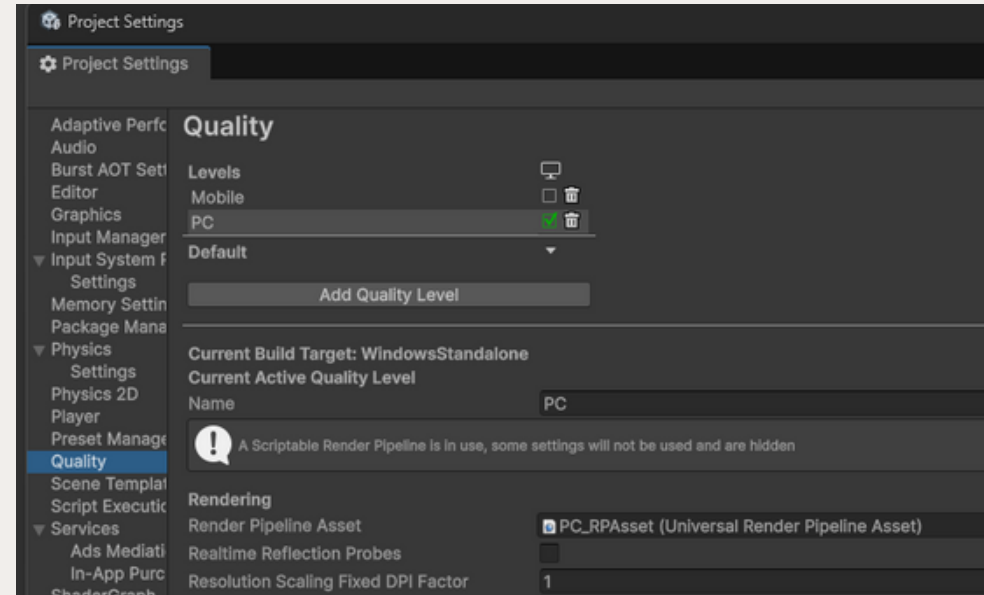
<URP에서 볼류메트릭 Cloud 생성하기>



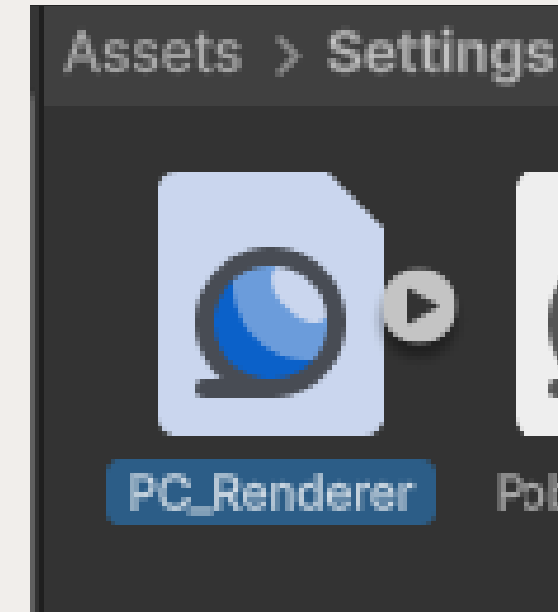
URP 프로젝트 생성



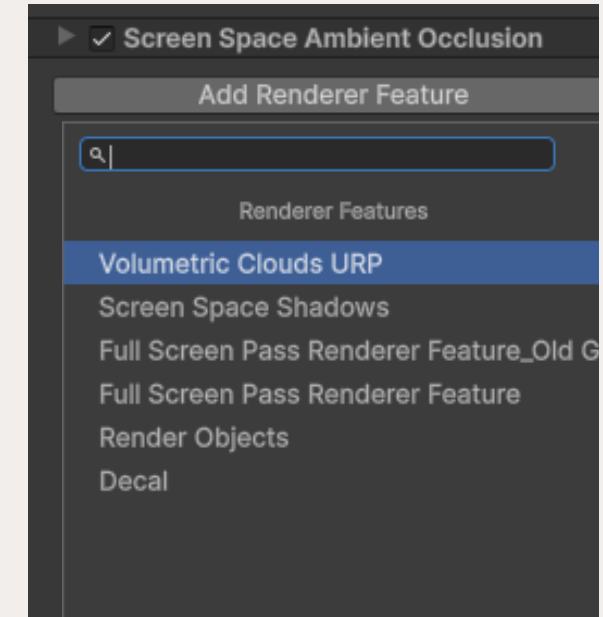
볼류메트릭 클라우드
에셋 다운로드



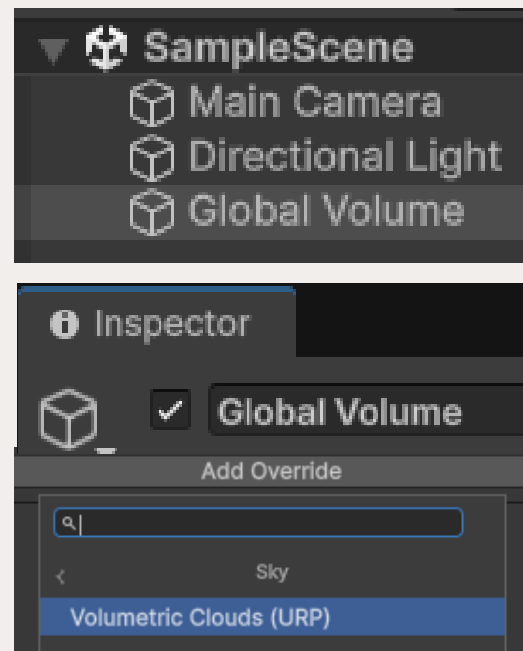
프로젝트 세팅 > Quality > Render
Pipeline Asset : URP 확인



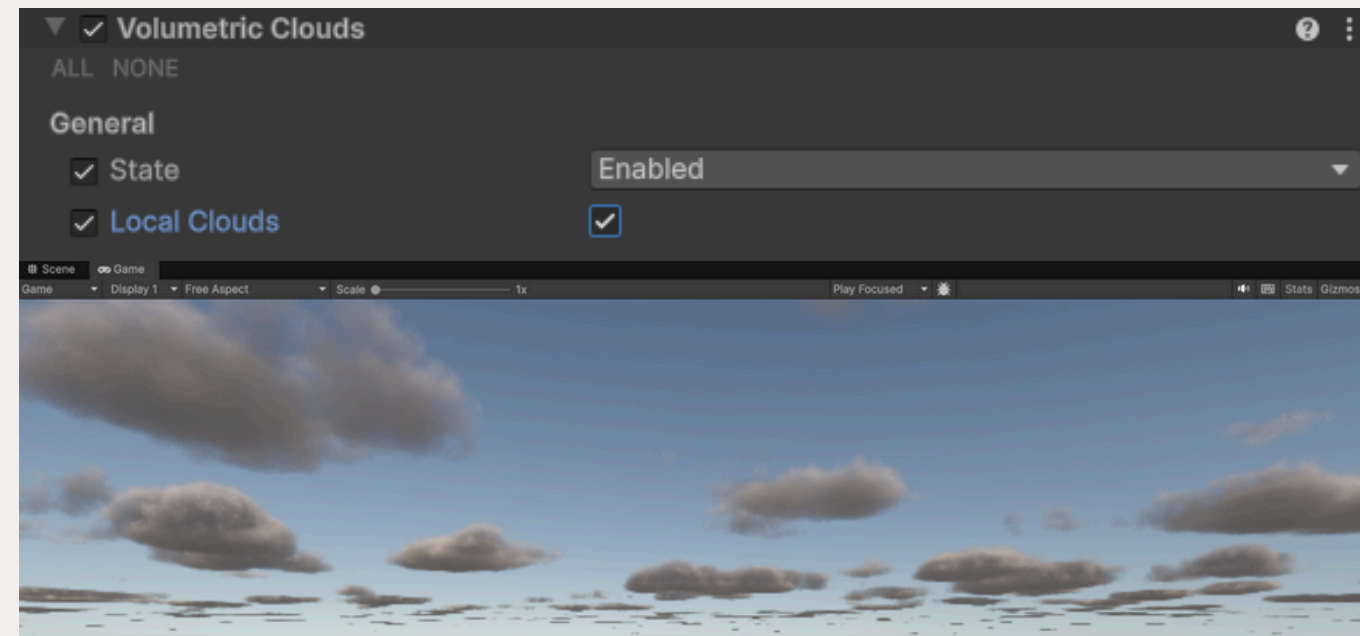
Asset>Setting
>PC_Renerer



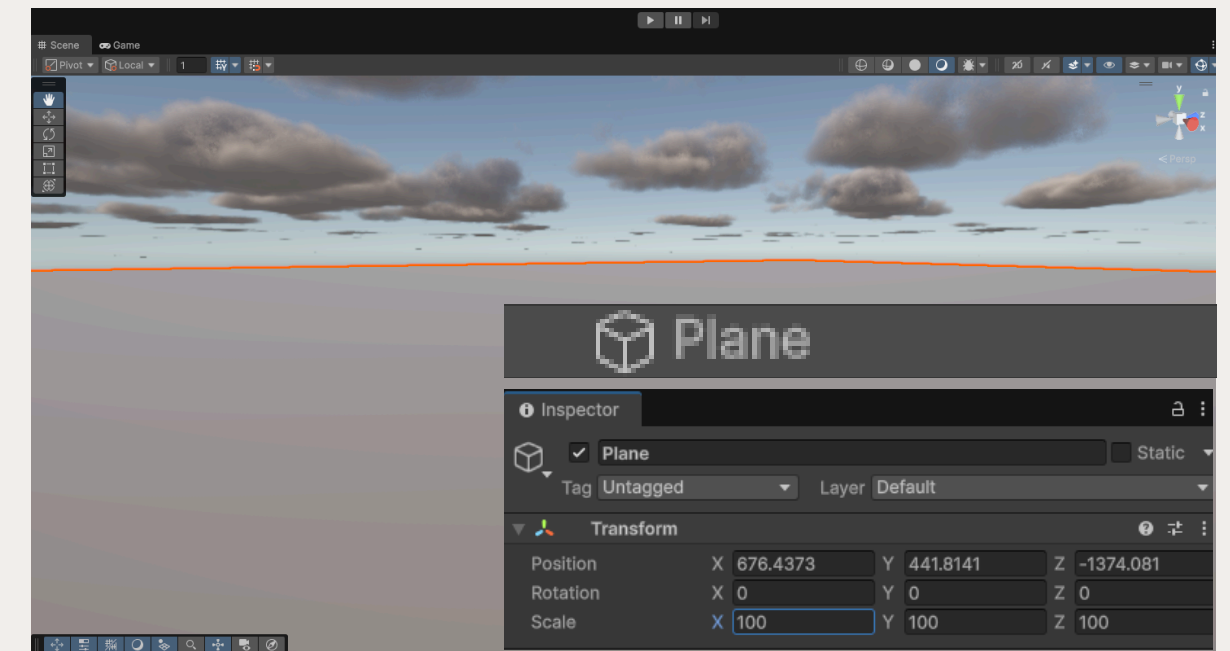
Add Renderer
Feature> V.C.URP



하이어라키>Global Volume
>인스펙터>AddOveeride
>Volumatric Clouds URP

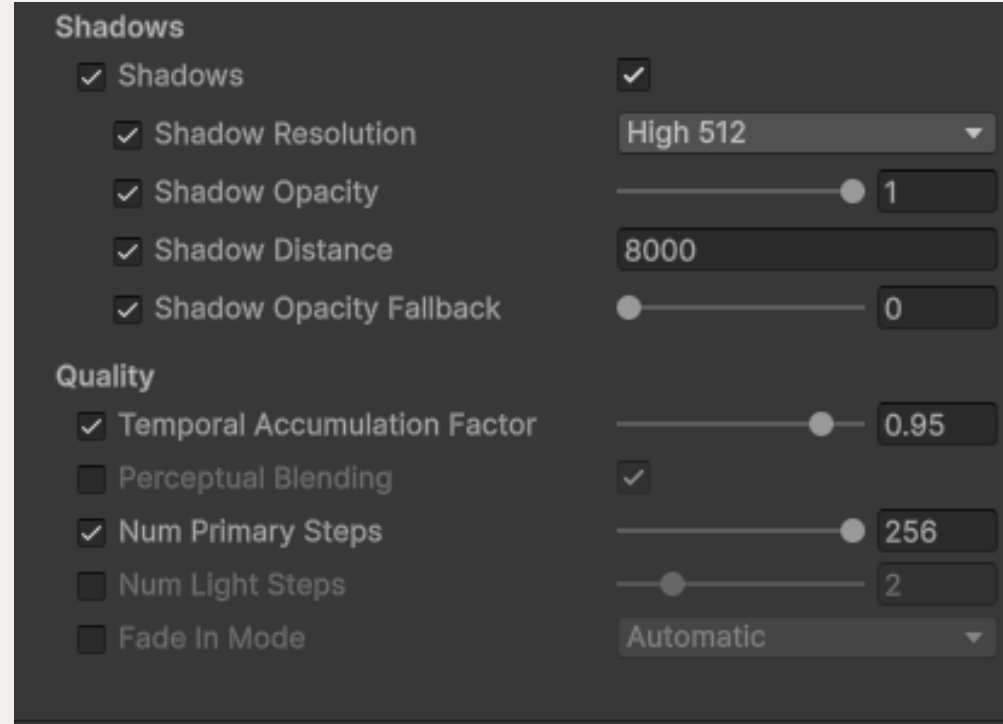


Volumatric Clouds > General:Enabled(활성화)
>Local Clouds [v]:실제 구름 에 접근 가능



Plane 생성 > 스케일 1000*1000*0100으로 확장

<URP에서 볼류메트릭 Cloud 생성하기>

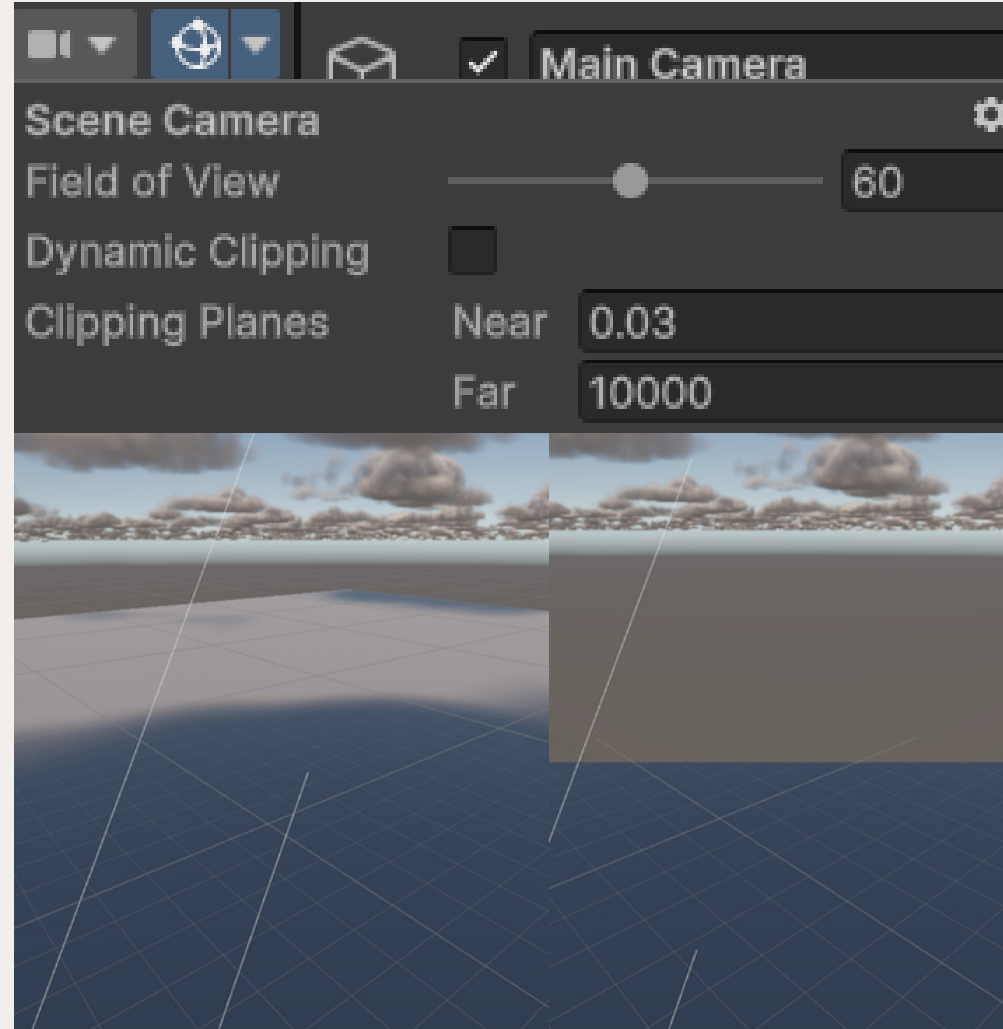


Shadows 활성화

- >Shadow Resolution(높을수록 고품질)
- >Shadow Opacity(그림자 진하기0~1)
- >Shadow Distance(그림자 보이는 거리)
- >Shadow Opacity Fallback(0~1)
(저사양 환경 그림자 대체 옵션 활성화)

Quality

- >Temporal Accumulation Factor (0.95)
(노이즈 줄임 없음0~1노이즈 감소 최대치)
- >Perceptual Blending(깜빡임 감)
- >Num Primary Steps(그림자 샘플링256)
- >Num Light Steps(빛계산 정밀도 2~)
- >Fade In Mode(그림자 거리 연산,Auto)



SceneView 카메라 클리핑 제한 해제:

카메라 옵션 > Dynamic Clipping []

- Near:0.1
- Far:10000

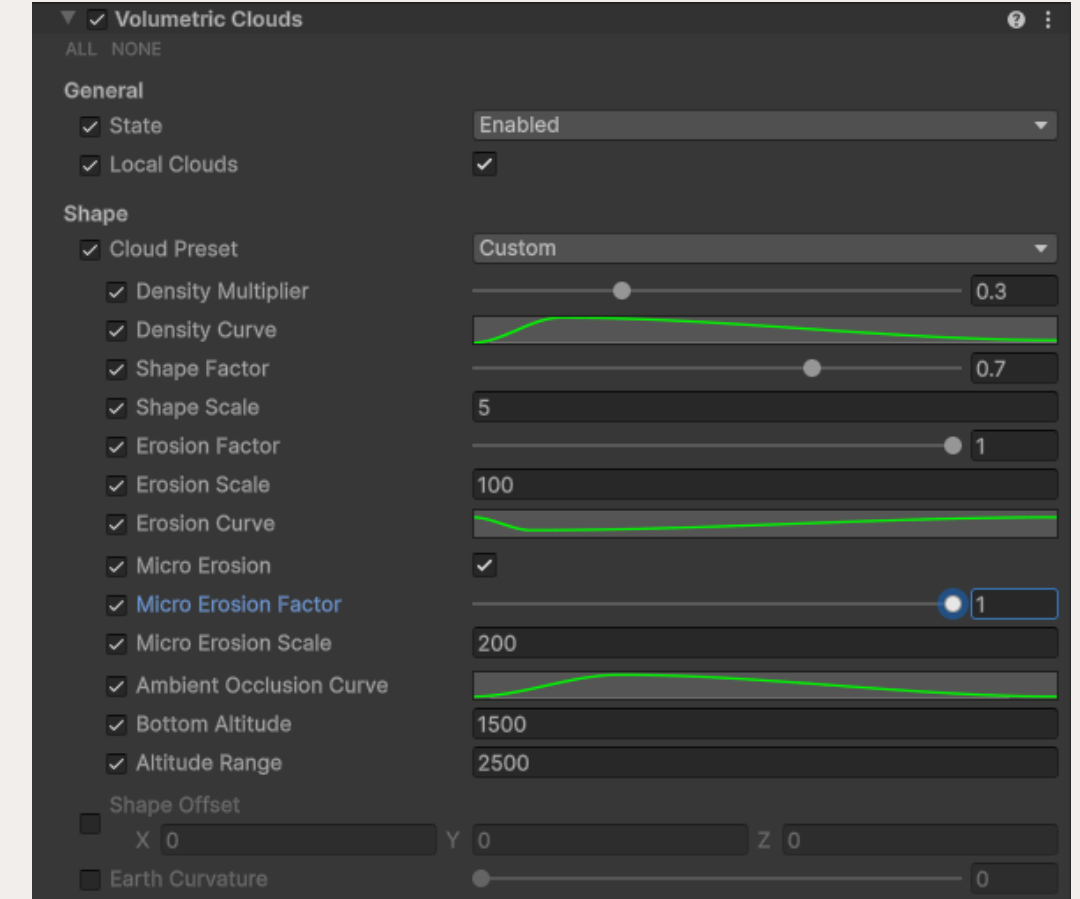
> SceneView 카메라 시야 확보

GameView 카메라 클리핑 제한 해제:

Main Camera > Clipping Planes>

- Near:0.1
- Far:10000

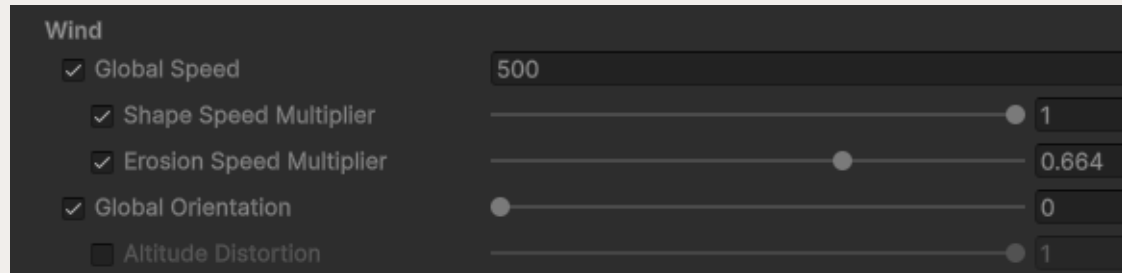
>GameView 카메라 시야 확보



Shape

- >Cloud Preset: 구름의 종류 프리셋
- >Density Multiplier: 밀도
- >Shape Factor: 구름 한 덩어리 크기
- >Shape Scale: 구름 크기
(낮으면:뭉게구름/높으면:양털구름)
- >Erosion Factor: 구름 가장자리 깎임 정도
- >Erosion Scale: 깎이는 패턴
- >Micro Erosion[v]>Factor:구름 투영도
- >Bottom Altitude:구름 고도
- >Altitude Range:구름 두께

<URP에서 볼류매트릭 Cloud 생성하기>



Wind

- >Global Speed: 구름 이동 속도
- >Shape Speed Multiplier: 구름 모양 변화 속도
- >Erosion Speed Multiplier: 가장자리 모양 변화 속도
- >Global Orientation: 월드 기준 구름 방향(0~360)
- >Altitude Distortion: 높이에 따라 구름 모양 왜곡
(ON → 구름이 더 자연스러워짐)

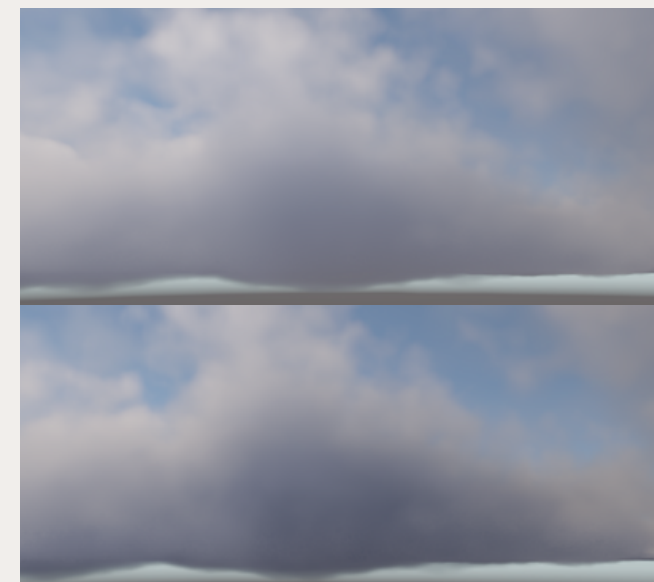


Lighting

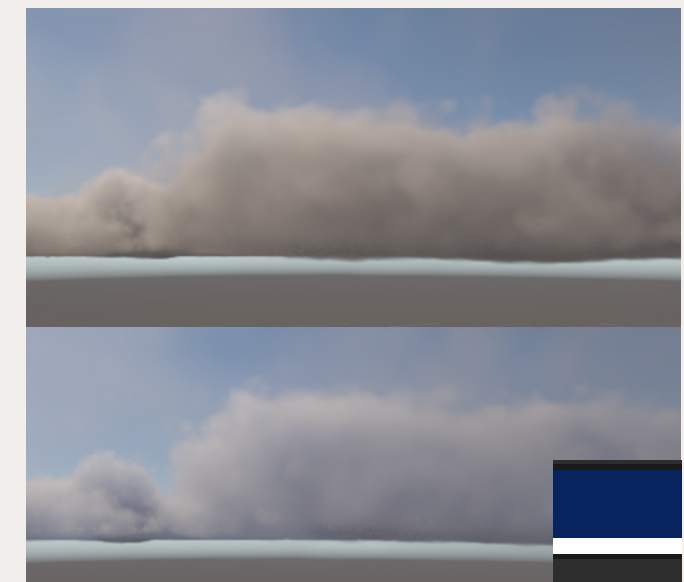
- >Ambient Light Probe Dimmer: 환경광 영향
- >Sun Light Dimmer: 태양 밝기 영향
- >Erosion Occlusion: 구름 내부 그림자 강도
- >Scattering Tint: 구름 색감
- >Powder Effect Intensity: 햇빛 부드러운 확산 느낌
- >Multi Scattering: 빛이 구름 안에서 여러 번 반사



Sun Light Dimmer

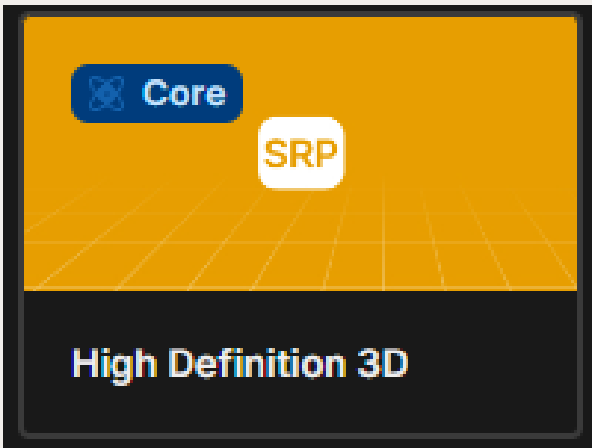


Erosion Occlusion

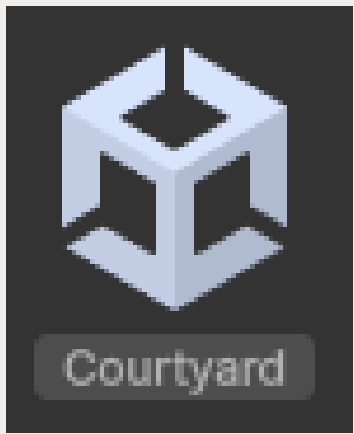


Scattering Tint

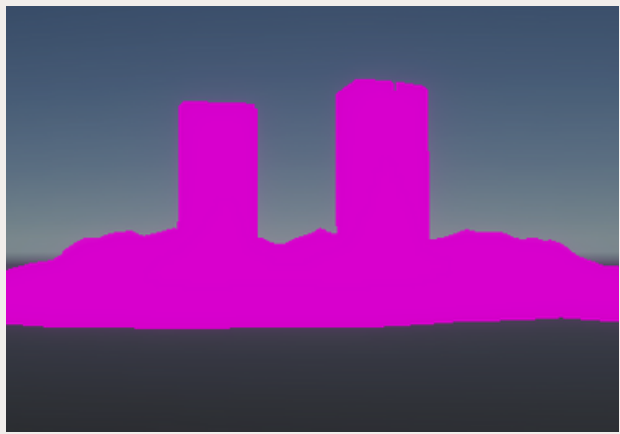
<HDRP에서 Realistic Lighting 설정방법>



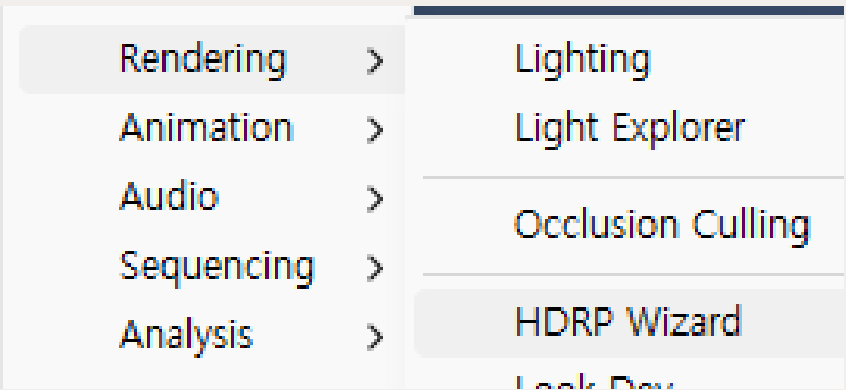
HDRP 프로젝트
생성



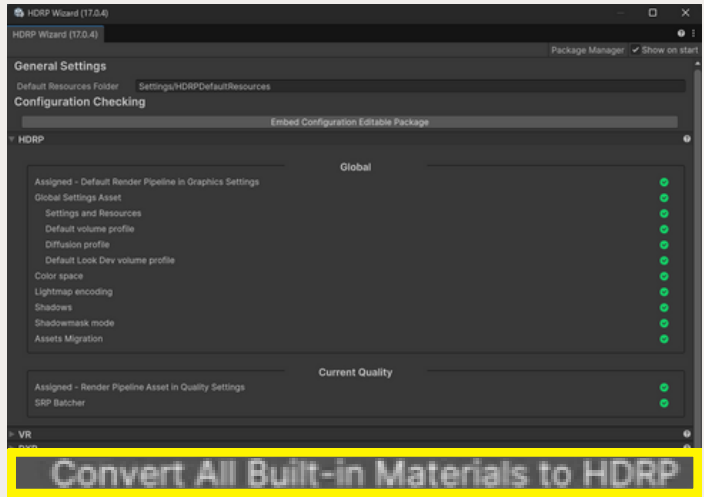
H.D.R.P.에셋
임포트



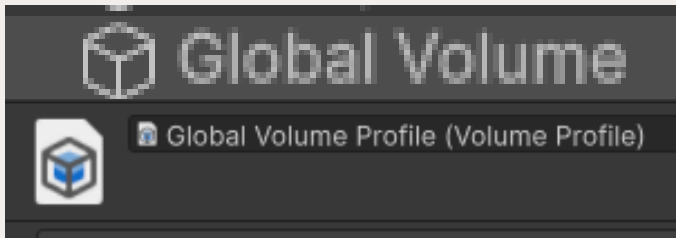
샘플 씬 열기
(County Yard)



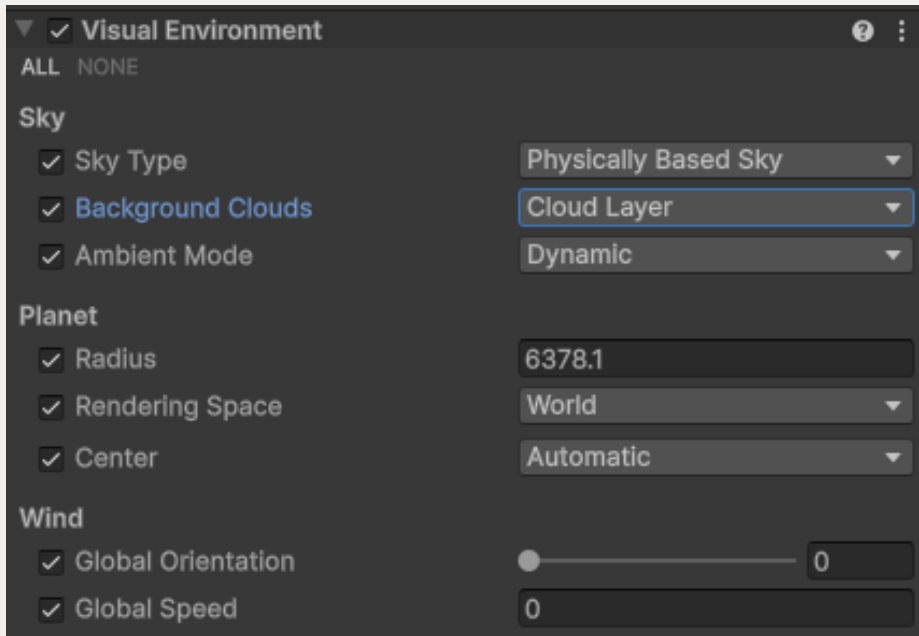
셰이더 오류 해결 위해
HDRP Wizard 열기



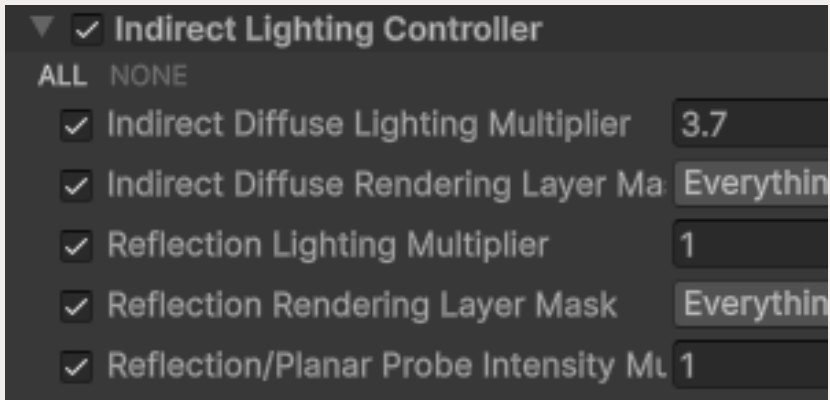
모든 머티리얼을
HDRP로 변환



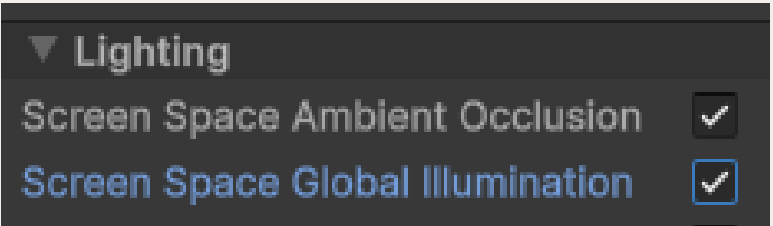
Global Volume 생성
>PPV에셋 생성



Add Override
>Visual Environment 생성
>All활성화>SkyType:PhysicalBasedSky
>BackgroundCloud:CloudLayer
>AmbientMode:Dynamic

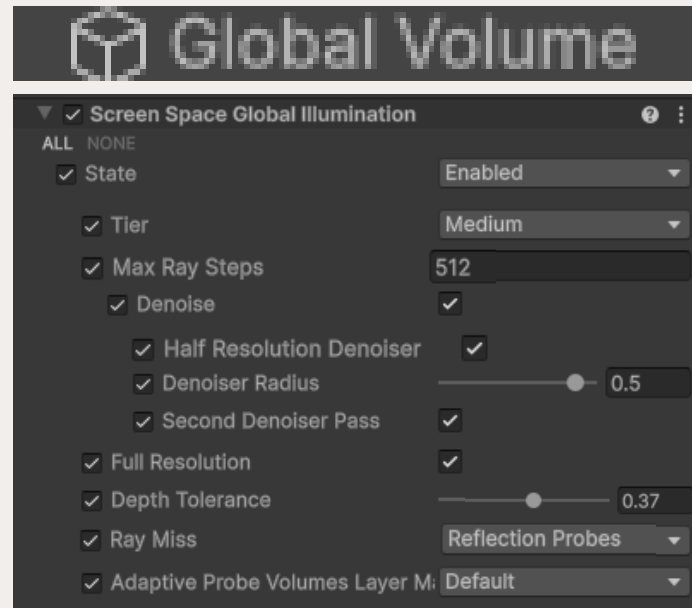


Add Override
>Indirect Lighting
Controller생성
> Indirect Diffuse
Lighting Multiplier:3

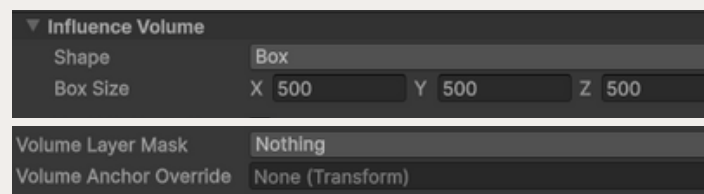


HDRP Asset > Lighting
>SSGI [v]

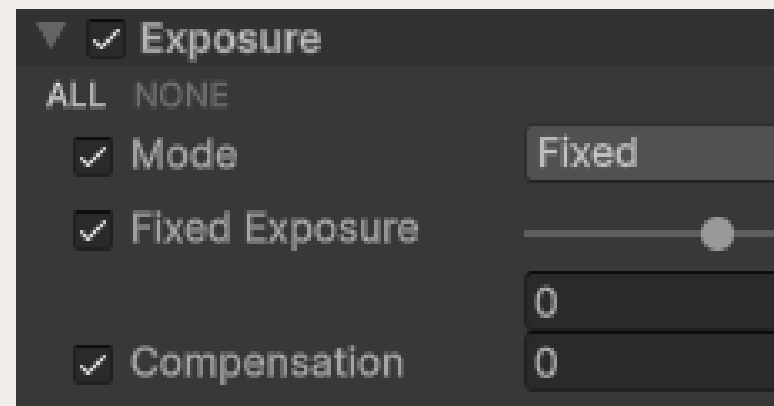
<HDRP에서 Realistic Lighting 설정방법>



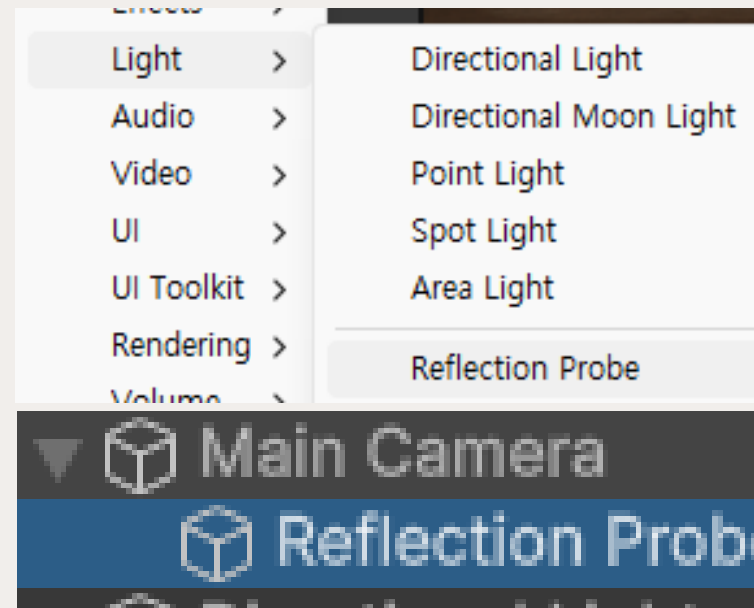
Global Volume>Add Override
 >SSGI추가>All활성 >State:Enabled
 >Max Ray Steps:레이트레이싱 성능.512
 >Half Resolution Denoiser:[v]
 (사용하는 GPU 고사양이 아닐시)
 >Depth Tolerance:0.37
 (깊이감 측정 정밀도,높으면 정밀)
 >Ray Miss: Reflection Probe
 (레이트레이싱 실패시 Reflection Probe
 환경으로 대체)



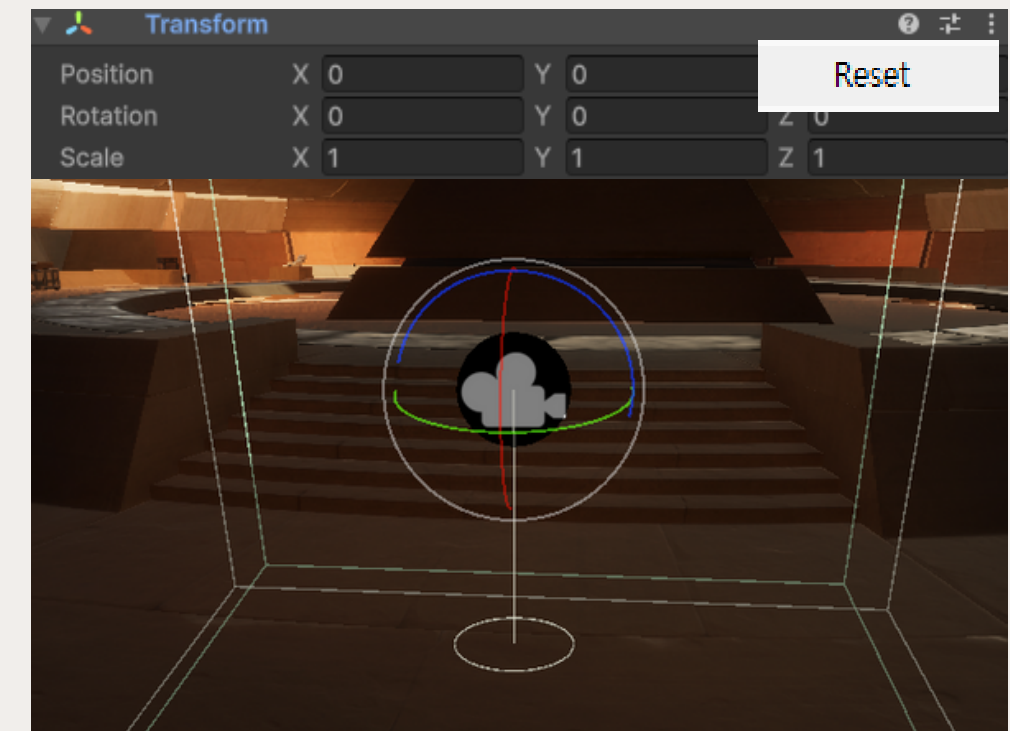
Reflection Probe> 적용 범위 확장
 >Volume Layer Mask:Nothing
 (어떤 Volume 데이터도 참조하지 않는다)
 *모든 광원들을 참조하면 너무 밝아지고,
 다중 연산으로 메모리에 과부하.



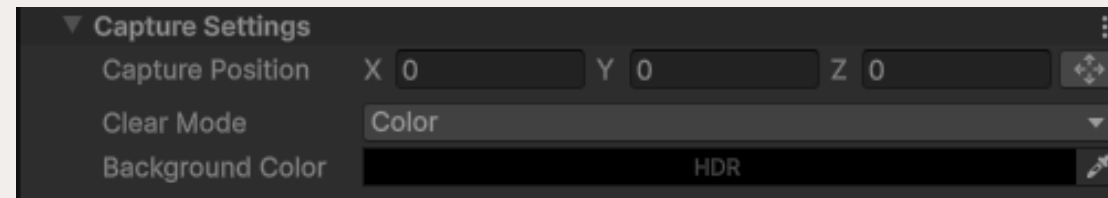
Add Override >Exposure추가
 >All활성화>Mode:Fixed
 >Fixed Exposure: 원하는 밝기



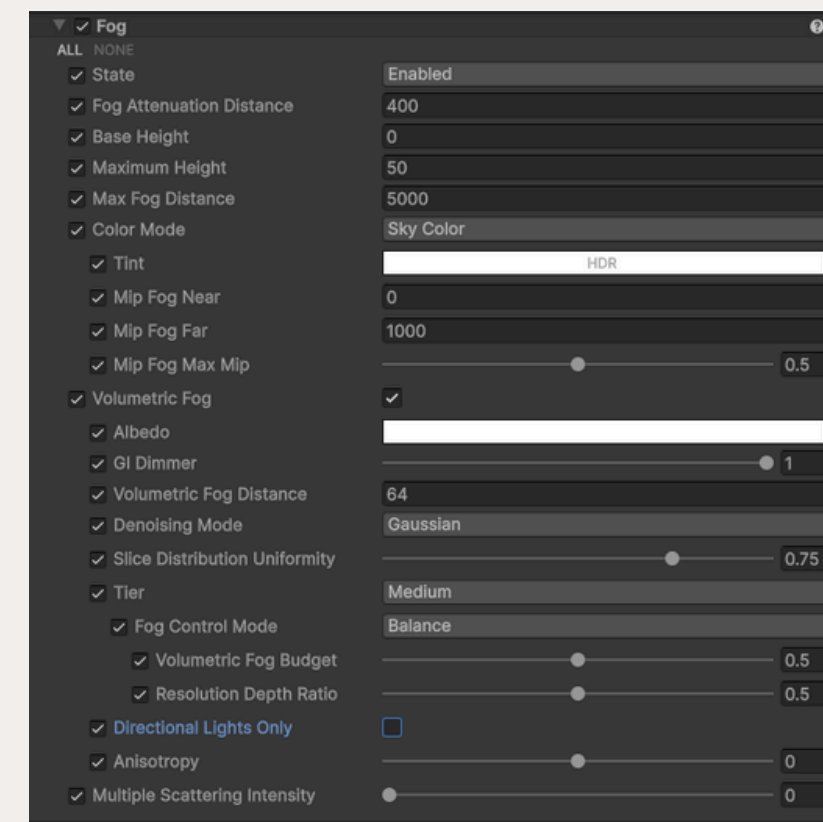
Hierarchy > 우클릭 > Light
 > Reflection Probe 추가
 >MainCamera 하위로 종속



ReflectionProbe의 트랜스폼 Reset
 하여 카메라와 동일 위치 배



Reflection Probe> Capture Setting >
 Clear Mode:Color > Background: Black
 (Reflection Probe가 캡처 전에 배경을 검정
 색으로 초기화→과한 양의 빛 캡처 방지)



Fog 추가